How-To Library

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Implementing PowerExchange Express CDC for Oracle in a Non-RAC Environment

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Abstract

This article describes how to implement PowerExchange 10.4.0 Express CDC for Oracle in a non-RAC Oracle environment on UNIX. It identifies the configuration steps that you perform in Oracle, PowerExchange, and PowerCenter to create a working change data capture (CDC) environment. It also includes example configuration files, tips, and best practices.

Supported Versions

- PowerExchange 10.4.0

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Overview

This article describes the steps that are necessary to successfully use PowerExchange Express CDC for Oracle to capture change data from Oracle tables in a non-RAC environment and use PowerCenter to apply the captured change data to target tables. The article includes example configuration files and best practices.

PowerExchange Express CDC for Oracle captures change data directly from Oracle active and archived redo logs or from copies of the archived redo logs and makes that data available to PowerCenter CDC sessions for propagation to targets. PowerExchange Express CDC is a much faster alternative to PowerExchange Oracle CDC with LogMiner. Express CDC produces a change stream in which inserts, updates, and deletes are organized by transaction, and the transactions are arranged in their original commit order.

Important: You cannot use both PowerExchange Express CDC for Oracle and PowerExchange Oracle CDC with LogMiner in the same PowerExchange instance with the same dbmover.cfg and pwxccl.cfg configuration files.

For PowerExchange Express CDC for Oracle, Informatica strongly recommends that you use of the PowerExchange Logger for Linux, UNIX, and Windows. The PowerExchange Logger gets the captured change data from Express CDC log reader and writes only the successful units of work (UOWs), in chronological order based on commit time, to PowerExchange Logger log files. PowerCenter CDC sessions, in conjunction with the PowerExchange Client for PowerCenter (PWXPC), can then extract the change data from the PowerExchange Logger log files in continuous extraction mode and apply the change data to target tables in near real time. Using the PowerExchange Logger has the following benefits:

- Fewer database accesses to read change data
- Faster CDC restart
- No need to prolong retention of Oracle redo files for CDC

Scenario

You need to keep a data warehouse synchronized with an Oracle production database in near real time.

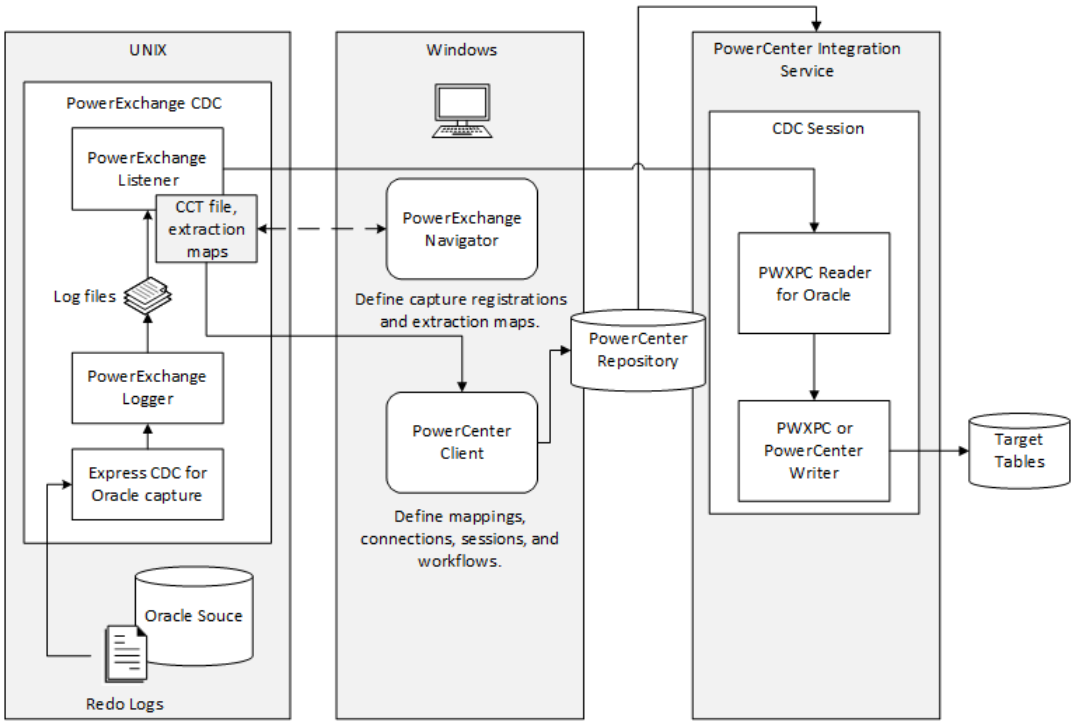
Your CDC environment has the following characteristics and requirements:

- You use PowerExchange 10.4.0 Express CDC for Oracle.
- You have an Oracle 12*c* R2 source database on a UNIX machine. The UNIX machine is remote from the PowerCenter Integration Service machine.
- You need to capture changes from about 50 Oracle source tables and move the change data in near real time to multiple target tables on a remote machine.
- The Oracle source tables are not in an Oracle Real Application Cluster (RAC), Oracle ASM environment, Oracle Data Guard standby environment, or Oracle 12*c* R2 multitenant environment.
- The PowerExchange Navigator and PowerCenter Client reside on the same Windows machine, which is remote from the PowerCenter Integration Service machine and Oracle source system.
- The PowerExchange Logger for Linux, UNIX, and Windows runs continuously on the Oracle source machine and logs change data records to local Logger log files.
- You extract change data from the PowerExchange Logger log files in *continuous extraction mode*. Continuous extraction mode provides near real time access to change data but is much less resource intensive than *real-time extraction mode*.

- You do not use offload processing. The Oracle source machine has sufficient memory and resources to handle CDC efficiently.
- You use PWXPC connections and PowerCenter workflows to extract the change data from the PowerExchange Logger log files and to write the data to multiple target tables.

  Note: PWXPC is a native plug-in that is installed with PowerCenter on the Windows and PowerCenter Integration Service machines.

  The following image shows the architecture for this scenario:



  The UNIX system hosts the Oracle source database, redo logs, and a PowerExchange instance, including the PowerExchange Logger, PowerExchange Listener, CCT file, and extraction maps. The Windows system hosts the PowerExchange Navigator and PowerCenter Client. In the PowerExchange Navigator, you create capture registrations and extraction maps and perform database row tests on the extraction maps. The PowerExchange Navigator reads and updates the CCT file and extractions maps on a UNIX source system. In the PowerCenter Client, you create mappings, connections, sessions, and workflows, which are stored in the PowerCenter repository. The PowerCenter Integration Service machine hosts the PWXPC reader and writer and communicates with the PowerCenter repository, PowerExchange Listener, and target.

Before You Begin

Before you configure PowerExchange Express CDC for Oracle, perform the following tasks:

1. Install the PowerExchange software on the following machines:
   - The UNIX machine that contains the Oracle source database and tables
   - The Windows machine where the PowerExchange Navigator and PowerCenter Client run
   - The PowerCenter Integration Service machine

   For installation instructions, see the *PowerExchange Installation and Upgrade Guide*.

2. Verify that a PowerCenter version that is compatible with the PowerExchange version is installed. With PowerExchange 10.4.0, you can use PowerCenter 10.0, 10.1, 10.2.0, or 10.4.0. For more information, see "PowerExchange Interoperability with PowerCenter" in the *PowerExchange Installation and Upgrade Guide*.
3. Gather the information about your Oracle environment that is required for CDC configuration. See "Gather Information for CDC Configuration" on page 5.
4. Review CDC limitations and restrictions. See "Express CDC for Oracle Considerations" on page 8.
5. Create the target tables.

PowerExchange Installation Notes

PowerExchange installation on UNIX and Windows is all-inclusive. You do not install PowerExchange components separately.

On the UNIX machine that contains the Oracle source tables, extract files from the PowerExchange tar file for your UNIX operating system. Then run install.sh to perform the installation. After installation completes, set the following environment variables:

- Set the PWX_HOME environment variable to point to the directory with the PowerExchange run-time components. For example: PWX_HOME=/usr/PowerExchange10.4.0.
- Edit the PATH and library path environment variables to append the PowerExchange installation directory. For example: PATH=/usr/PowerExchange10.4.0:$PATH.
- Verify that the LANG and LC_ALL environment variables are set to "C" for ASCII. If these environment variables are not set to "C," ask your UNIX administrator to change them.
- (Recommended) Set the ORACLE_HOME and ORACLE_SID environment variables to bypass the use of SQL*Net for connection to the Oracle database. These variables improve PowerExchange Logger performance, even when the Logger is running on the Oracle source system. If you set these variables, you do not need to specify the fourth positional parameter, *capture_connnect_string*, in the ORACLEID statement in the dbmover.cfg file.

On the Windows machine, unzip the pwx*vrm*_win_x64.zip file and then run the install.bat file. This installation includes the 64-bit executables for the PowerExchange Navigator and the 32-bit executables required to support the PowerCenter Client. The installation program updates the Path environment variable to include the PowerExchange installation directory, which is required for the PowerExchange Navigator to work.

On the PowerCenter Integration Service machine, follow the PowerExchange installation instructions for the operating system type.

For more information about PowerExchange installation procedures, see the *PowerExchange Installation and Upgrade Guide*.

Gather Information for CDC Configuration

Gather the information that you will need to enter for Oracle CDC configuration.

For this scenario, assume that the PowerExchange installation is at */usr*/PowerExchange10.4.0. Also assume that you store the CCT file, extraction maps, message logs, and PowerExchange Logger log files in directories that are separate from the PowerExchange code, as recommended.

The following table identifies the information to gather and provides example values:

Information	Where Specified	Example Value	Value
Oracle Source Information:			
Oracle database name	Second positional parameter, *oracle_db*, in ORACLEID statements in the dbmover.cfg files on the UNIX and Windows machines	orad01	
Oracle source table names in the format *schema.table*, and the table columns for which to capture change data	Capture registrations that you create from the PowerExchange Navigator	-	
The Oracle archive log destination	Oracle init.ora file or spfile customization SQL	/*oracle_path*/arch	
An Oracle CDC user name	The SQL CREATE USER and GRANT statements that you will execute to create a CDC user and grant the required Oracle privileges to this user	ORACAPTL1	
A user-defined collection ID value for the Oracle source instance. Maximum length is eight characters.	The following statements in the dbmover.cfg file on the UNIX machine: - First positional parameter in the ORACLEID statement - ORACOLL parameter in the ORAD CAPI_CONNECTION - DFLTINST parameter in the CAPX CAPI_CONNECTION Also specified in: - The **Collection ID** field that you set when you create a registration group in the PowerExchange Navigator, or the **Instance** field that you view in registration group properties - DBID parameter in the pwxccl.cfg file	ORAD1DB	
An Oracle connection string defined in TNS that the PowerExchange Navigator will use to access the Oracle source database	Third positional parameter in the ORACLEID statements in the DBMOVER configuration files on the UNIX and Windows machines. This value is a connect string in the Oracle Client tnsnames.ora file on the UNIX machine.	orad01.world	
LISTENER and NODE Statements:			
A user-defined name for the LISTENER statement on the UNIX machine, which will be used to start or stop the Listener	The LISTENER statement in the dbmover.cfg file on the UNIX machine	pwxlst	

Information	Where Specified	Example Value	Value
The port number on which the PowerExchange Listener on the UNIX machine listens	The LISTENER statement in the DBMOVER configuration file on the UNIX machine The NODE statements in the DBMOVER configuration files on the following machines: - The Windows machine that contains the PowerExchange Navigator and PowerCenter Client - The PowerCenter Integration Service machine	2480 **Note:** This port number is the default value.	
A user-defined node name for the NODE statement that will be used to connect to the UNIX machine from the Windows or PowerCenter Integration Service machine	The NODE statements in the DBMOVER configuration files on the following machines: - The Windows machine that contains the PowerExchange Navigator and PowerCenter Client - The PowerCenter Integration Service machine	ora_unix1	
The TCP/IP host name or IP address of the UNIX machine	The NODE statements in the DBMOVER configuration files on the following machines: - The Windows machine where the PowerExchange Navigator and PowerCenter Client run - The PowerCenter Integration Service machine	unix_host1	
File Locations and Retention:			
Path to PowerExchange message log files on the UNIX machine	LOGPATH statement in the DBMOVER configuration file on the UNIX machine	/Informatica/PowerExchange10.4.0/capture/logs	
Subdirectory of the LOGPATH directory for alternative message log files for the PowerExchange Logger (optional)	PFX option of the TRACING statement in the dbmover.cfg file on the UNIX machine	altlogs	
Path and unique prefix for the generated PowerExchange Logger log files. To create the log file names, the Logger appends the following values: `.CND.CPyymmdd.Thhmmssnnn`	EXT_CAPT_MASK statement in the pwxccl.cfg file on the UNIX machine	/capture/pwxccl_log	

Information	Where Specified	Example Value	Value
Retention period, in days, for CDCT records and PowerExchange Logger log files	COND_CDCT_RET_P statement in the pwxccl.cfg file on the UNIX machine When setting this parameter, try to minimize the size of the CDCT file while preserving the log files that contain the earliest change data that you might need to access. In continuous extraction mode, PowerExchange reads the CDCT file each time the interval specified in the FILEWAIT parameter of the CAPX CAPI_CONNECTION statement elapses. If a CDCT file becomes large, this read activity can increase I/O, system resource use, and latency of change data extraction.	60 Default is 60.	
Path to the directory that stores the CDC control files on the UNIX machine, including the CCT file for capture registrations and the CDCT file for PowerExchange Logger log files and restart information	CAPT_PATH statement in the dbmover.cfg file on the UNIX machine	/Informatica/PowerExchange10.4.0/capture	
Path to the directory that stores extraction maps on the UNIX machine	CAPT_XTRA statement in the dbmover.cfg file on the UNIX machine	/Informatica/PowerExchange10.4.0/capture/xtracmaps	

Express CDC for Oracle Considerations

Consider the following requirements, restrictions, and operational considerations that pertain to Oracle CDC in this scenario:

- Oracle ARCHIVELOG mode and minimal global supplemental logging must be enabled. Also, the registered Oracle source table columns for which you want to capture change data must be associated with an unconditional supplemental log group, also called an ALWAYS supplemental log group. With unconditional supplemental logging, Oracle logs before images of column data to redo logs whenever any column in a row is updated. PowerExchange Express CDC for Oracle requires before images to properly process updates.
- You cannot run both PowerExchange Express CDC for Oracle and PowerExchange Oracle CDC with LogMiner with the same PowerExchange instance and the same PowerExchange Listener and PowerExchange Logger. If you need to use both of these Oracle CDC solutions, run each one in a separate PowerExchange instance with different dbmover.cfg and pwxccl.cfg configuration files. However, you can use the same capture registrations for both CDC solutions if you want to capture change data from the same Oracle database and tables. For these CDC solutions to share registration information, store the CCT file in a common location.
- The PowerExchange Express CDC for Oracle log reader must have access to the Oracle redo log files. When the Oracle redo log files are *not* stored in ASM, as in this scenario, the operating system user ID under which PowerExchange Express CDC for Oracle log reader runs must have the authority to read the redo logs or copies of the archived redo logs.
- PowerExchange Express CDC for Oracles reads the active and archived redo logs by default. If you want Express CDC to read only the archived redo logs, set the MODE parameter to ARCHIVEONLY in the Express CDC configuration file. If you do not have the authority to read the active or archived redo logs, you can

configure Express CDC to read copies of the archived logs that are located on a file system. In this case, set the MODE parameter to ARCHIVECOPY.

- If you capture change data only from archived redo logs and use the PowerExchange Logger, the default restart point after a PowerExchange Logger cold start is the end of the last available archived log, also referred to as current end-of log (EOL), unless you set the RESTART_TOKEN and SEQUENCE_TOKEN parameters in the pwxccl.cfg file.
- You must use the PowerExchange Client for PowerCenter (PWXPC) to integrate with PowerCenter. The PowerExchange ODBC driver does not support PowerExchange Express CDC for Oracle.
- PowerExchange Express CDC for Oracle does not support the following Oracle environments, objects, and features:
  - Any type of encryption of tablespaces other than Oracle Advanced Security Transparent Data Encryption (TDE) of tablespaces
  - Tables in a sorted hash cluster
  - Columns that use TDE
  - Virtual columns, which contain derived data that Oracle does not log in the redo logs
  - Columns that have unsupported datatypes
    Because you cannot include these columns in capture registrations, PowerExchange Express CDC for Oracle does not capture change data for them. However, PowerExchange Express CDC for Oracle can capture change data for other columns in the same registered table.
- If a source table contains LOB columns that are defined with the STORE AS SECUREFILE storage parameter, PowerExchange Express CDC for Oracle cannot capture change data from these LOB columns but can capture change data from the other columns in the table.
- PowerExchange Express CDC for Oracle can capture data that the SQL*Loader utility loaded into Oracle tables provided that the load method is Insert, Append, or Replace.
- If you use Oracle materialized views, PowerExchange can capture change data from the master tables that underlie those views. PowerExchange supports change capture for any type of materialized view. The view and its underlying table have a one-to-one correspondence and share the same name.

  If you issue DTLDESCRIBE tables from the Database Row Test dialog box in the PowerExchange Navigator, the results include a row for the materialized view and a row for the underlying table. The Type column indicates which row is for the materialized view and which row is for the table.
- PowerExchange Express CDC for Oracle uses the standard PowerExchange restart and recovery processing for relational sources. The format of PowerExchange Express CDC for Oracle restart tokens is different from that for Oracle CDC with LogMiner sources or for any other data source type.

  The default restart point for PowerExchange Express CDC for Oracle is one of the following points in the change stream:
  - For CDC sessions that run in batch extraction mode or continuous extraction mode, the beginning of the oldest PowerExchange Logger log file that is recorded in the CDCT file.
  - For CDC sessions that run in real-time extraction mode, the beginning of last log sequence that was archived.
- For tables that do not have row movement enabled, you can use the PowerExchange-generated DTL__CAPXROWID column in extraction maps and the OPTIONS ROWID=Y statement in the PowerExchange Express CDC for Oracle configuration file to include Oracle physical rowid values in change records. This feature is useful for processing rows in unkeyed tables during CDC extraction sessions.
- PowerExchange Express CDC for Oracle can capture Oracle direct-path operations except for tables that use Oracle Exadata Hybrid Columnar Compression (EHCC). To capture direct-path operations, you must set the SUPPORT_DIRECT_PATH_OPS parameter to Y in the OPTIONS statement of the PowerExchange Express CDC for Oracle configuration file.

- In Oracle 12*c* R2 and later environments, PowerExchange Express CDC for Oracle can capture conventional and direct-path DML changes from tables and table partitions and subpartitions that use Oracle Advanced Compression.
- In Oracle 12*c* R2 and later environments, PowerExchange Express CDC for Oracle can capture conventional DML changes from tables and table partitions and subpartitions that use Oracle Exadata Hybrid Columnar Compression (EHCC). However, Express CDC does not capture Oracle direct-path operations for objects that use EHCC.
- PowerExchange Express CDC for Oracle can capture change data from Oracle tablespaces that use Oracle Advanced Security Transparent Data Encryption (TDE). To capture TDE-encrypted change data from tablespaces, you must perform the following tasks:
  - Enter either the TDEWALLETPWD or TDEWALLETEPWD parameter in the pwxorad.cfg configuration file to specify the password for the Oracle TDE wallet.
  - Grant the following privilege to the ORACAPTL user:

    ```
    GRANT SELECT ON "PUBLIC"."V$ENCRYPTION_WALLET" TO "ORACAPTL";
    ```
  - Ensure that the Oracle TDE wallet is on a device that PowerExchange Express CDC for Oracle can access with Read file permissions.
  - Verify that the Oracle TDE wallet is open in the database.
- PowerExchange Express CDC can capture change from an Oracle database in an Amazon Elastic Compute Cloud (EC2) environment. The Amazon EC2 instances must run on a 64-bit Red Hat Linux server. Express CDC does not support EC2 instances on 64-bit Windows servers.
- If an Oracle RESETLOGS event occurs on a source database, PowerExchange Express CDC can continue change capture processing across the RESETLOGS boundary in the archived redo logs.
- If the memory that PowerExchange Express CDC for Oracle uses to stage change records becomes full, for example, because of large UOWs or a high transaction volume, PowerExchange Express CDC for Oracle can spill additional change records to a temporary spill file on disk. If CDC performance becomes degraded, increase the MEMOPS parameter value in the OPTIONS statement of the PowerExchange Express CDC for Oracle configuration file to a level that is sufficient for the change data volume.
- If you use continuous extraction mode, minimize the size of the CDCT file. The CDCT file contains information about PowerExchange Logger log files. PowerExchange reads the CDCT file each time the interval that is specified in the FILEWAIT parameter of the CAPX CAPI_CONNECTION statement elapses. If a CDCT file is large, PowerExchange read operations can result in a high level of I/O activity, increased use of system resources, and increased extraction latency. To manage the CDCT file size, use the COND_CDCT_RET_P statement in the pwxccl.cfg configuration file for the PowerExchange Logger.

Configure Oracle for Express CDC

You must perform some configuration tasks in Oracle to prepare for PowerExchange Express CDC for Oracle.

If you have not configured the use of archived logs, enable ARCHIVELOG mode and specify a primary archive log destination. Then use the sample file, ora_orad.sql, that PowerExchange supplies in its installation directory to perform the other tasks. To run the SQL statements in the ora_orad.sql file, you must be assigned the role of DBA.

Complete the following Oracle configuration tasks:

1. Specify an archive log destination for PowerExchange Express CDC for Oracle use if one is not already defined.
2. Enable ARCHIVELOG mode if it is not already enabled.
3. If you enable ARCHIVELOG mode, stop and restart the Oracle database.
4. Create an Oracle CDC user and grant the privileges required for CDC to this user.

5. Enable Oracle minimal global supplemental logging.

Specifying an Archive Log Destination

You must specify the archive log destination from which the Express CDC log reader selects archived redo log files.

Consult with your Oracle DBA. To configure the archived log destination, complete one of the following actions:

- Edit the init.ora file to specify the archive log destination and file-name format. For more information about the init.ora file, see the Oracle database administrator guide.
- Customize the appropriate server parameter file (spfile) to indicate the archived log destination, for example:

```
CONNECT SYS/sys_pwd AS SYSDBA;
     ALTER SYSTEM SET
     log_archive_dest_1 = 'location=/oracle_path/arch'
     SCOPE=SPFILE;
```

- For Oracle databases deployed in an Amazon RDS for Oracle environment, create the directories ARCHIVELOG_DIR and ONLINELOG_DIR for the archived and online redo logs, respectively. Use the following exec statements:

```
exec rdsadmin.rdsadmin_master_util.create_archivelog_dir;
exec rdsadmin.rdsadmin_master_util.create_onlinelog_dir;
```

  Also define an appropriate retention policy for the archived redo logs. Use the following exec statement:

```
exec rdsadmin.rdsadmin_util.set_configuration('archivelog retention days',number_of_days);
```

  Tip: The Express CDC for Oracle user will require read-only access to the online and archived redo log files in these directories.

Enabling ARCHIVELOG Mode

PowerExchange Express CDC for Oracle requires Oracle to be running in ARCHIVELOG mode.

By default, ARCHIVELOG mode is not enabled.

To enable ARCHIVELOG mode for Oracle databases not in an Amazon RDS environment, issue the following SQL statements:

```
SHUTDOWN IMMEDIATE;
STARTUP MOUNT;
ALTER DATABASE ARCHIVELOG;
ALTER DATABASE OPEN;
SHUTDOWN IMMEDIATE;
STARTUP;
```

Tip: Back up your database after each SHUTDOWN command.

To confirm that ARCHIVELOG mode is set for the database, enter the following SQL statement:

```
SELECT LOG_MODE FROM SYS.V&DATABASE;
```

For an Amazon RDS for Oracle databases, set the backup retention period to place the database in ARCHIVELOG mode and enable automated backups.

Creating an Oracle User and Granting User Privileges

To capture change data from Oracle redo logs, you must define a CDC user and assign specific Oracle system and object privileges to that user.

You can either use an existing user who has the required authority as the CDC user, or create a user and grant the required privileges to that user. In the ora_orad.sql file, PowerExchange provides sample SQL statements for creating an Oracle user and for granting the required privileges to that user.

1. Log in to Oracle as a user who has the role of DBA.
2. To create a user named ORACAPTL1, issue the following SQL statement:

```
CREATE USER "ORACAPTL1"  PROFILE "DEFAULT"
    IDENTIFIED BY "Oracaptl1"
    ACCOUNT UNLOCK;
COMMIT;
```

3. To retrieve the information that is required for correct capture processing, grant the following minimum object and system table privileges to the ORACAPTL1 user:

```
GRANT SELECT ON "PUBLIC"."V$ARCHIVED_LOG"        TO "ORACAPTL1";
GRANT SELECT ON "PUBLIC"."V$DATABASE"               TO "ORACAPTL1";
GRANT SELECT ON "PUBLIC"."V$LOG"                    TO "ORACAPTL1";
GRANT SELECT ON "PUBLIC"."V$LOGFILE"                TO "ORACAPTL1";
GRANT SELECT ON "PUBLIC"."V$TRANSPORTABLE_PLATFORM" TO "ORACAPTL1";
GRANT SELECT ON "PUBLIC"."V$THREAD"                 TO "ORACAPTL1";
GRANT SELECT ON "PUBLIC"."V$DATABASE_INCARNATION"   TO "ORACAPTL1";
GRANT SELECT ON "PUBLIC"."V$PARAMETER"              TO "ORACAPTL1";
GRANT SELECT ON "PUBLIC"."V$SPPARAMETER"            TO "ORACAPTL1";
GRANT SELECT ON "PUBLIC"."V$STANDBY_LOG"            TO "ORACAPTL1";

GRANT SELECT ON "SYS"."ALL_TABLES"         TO "ORACAPTL1";
GRANT SELECT ON "SYS"."ALL_TAB_PARTITIONS" TO "ORACAPTL1";

GRANT SELECT ON "SYS"."DBA_LOG_GROUPS"        TO "ORACAPTL1";
GRANT SELECT ON "SYS"."DBA_LOG_GROUP_COLUMNS" TO "ORACAPTL1";
GRANT SELECT ON "SYS"."DBA_USERS"             TO "ORACAPTL1";
GRANT SELECT ON "SYS"."DBA_TABLESPACES"       TO "ORACAPTL1";

GRANT SELECT ON "SYS"."OBJ$"          TO "ORACAPTL1";
GRANT SELECT ON "SYS"."TAB$"          TO "ORACAPTL1";
GRANT SELECT ON "SYS"."IND$"          TO "ORACAPTL1";
GRANT SELECT ON "SYS"."COL$"          TO "ORACAPTL1";

GRANT SELECT ON "SYS"."PARTOBJ$"      TO "ORACAPTL1";
GRANT SELECT ON "SYS"."TABPART$"      TO "ORACAPTL1";
GRANT SELECT ON "SYS"."INDPART$"      TO "ORACAPTL1";
GRANT SELECT ON "SYS"."TABCOMPART$"   TO "ORACAPTL1";
GRANT SELECT ON "SYS"."TABSUBPART$"   TO "ORACAPTL1";
COMMIT;
```

4. To extract change data in real time and run PowerExchange Logger tasks, grant the following system privilege to the ORACAPTL1 user:

```
GRANT CREATE SESSION TO "ORACAPTL1";
```

5. To capture data from Oracle TDE-encrypted tablespaces, grant the following privilege to the ORACAPTL1 user:

```
GRANT SELECT ON "PUBLIC"."V$ENCRYPTION_WALLET" TO "ORACAPTL1";
```

6. To create capture registrations and perform other tasks in the PowerExchange Navigator, grant the following privileges to the user who creates and manages registrations:

```
GRANT SELECT ON "PUBLIC"."V$PARAMETER" TO "registration_user";
GRANT SELECT ON table TO "registration_user";  <<-Repeat for each table of CDC interest.
```

Instead of granting SELECT on each table of interest, you can specify `GRANT SELECT ANY TABLE` for the *registration user* if your site security rules allow it.

7. To run the SQL for creating supplemental log groups at the end of registration, grant the following system privilege to the user who created the registration:

   ```
   GRANT ALTER ANY TABLE TO "registration_user";
   ```

   If your site security rules do not allow this level of authority to be granted to the registration user, you can give the SQL file to your DBA. The DBA can then use the SQL to create the supplemental log groups.

Attention: In a non-ASM environment, the operating system user ID under which the PowerExchange Express CDC for Oracle capture process runs must also have the authority to read the Oracle online and archive redo logs. Otherwise, the Express CDC log reader cannot directly read the log files, and the CDC session ends with error message PWX-36140.

Enabling Oracle Minimal Global Supplemental Logging

PowerExchange Express CDC for Oracle requires Oracle minimal global supplemental logging to properly handle chained rows.

To enable supplemental logging for an on-premises database, log in to the Oracle database and issue the following SQL statement, which is included in the sample ora_orad.sql script file:

```
ALTER DATABASE ADD SUPPLEMENTAL LOG DATA;
COMMIT;
```

If you execute this statement when minimal global supplemental logging is already enabled for your database, the statement has no effect.

If you execute this statement while the Oracle database is open, Oracle waits for in-flight transactions to complete, which can affect database performance. This issue is likely to occur for databases that have a high level of user activity. To avoid this problem, you can close and re-open the database and then issue the statement manually.

To enable supplemental logging at the database level in an Amazon RDS for Oracle environment, execute the following exec statement:

```
exec rdsadmin.rdsadmin_util.alter_supplemental_logging('ADD');
```

Note: You must also define a supplemental log group for each Oracle source table. When you register an Oracle table in the PowerExchange Navigator, PowerExchange generates DDL for adding a supplemental log group for the table. Supplemental log groups cause Oracle to log full before- and after-images of the data that changed. PowerExchange requires these images.

Configure PowerExchange for Express CDC

You must perform several tasks to configure PowerExchange for Express CDC for Oracle.

1. On the UNIX system where the Oracle database, PowerExchange capture process, and PowerExchange Logger run, perform the following configuration tasks:
   - Customize the PowerExchange Express CDC for Oracle configuration file, pwxorad.cfg.
   - Customize the PowerExchange Logger configuration file, pwxccl.cfg.
   - Configure the dbmover configuration file.
2. On the Windows system where the PowerExchange Navigator runs, perform the following configuration tasks:
   a. Configure the dbmover configuration file.
   b. In the PowerExchange Navigator, create a registration group for the Oracle source instance. Then add a capture registration for each source table in the group. The PowerExchange Navigator generates extraction maps that correspond to the added capture registrations.

c. Perform a database row test on each generated extraction map.

3. On the PowerCenter Integration Service system, configure the dbmover configuration file.

Customizing the PowerExchange Express CDC for Oracle Configuration File

To specify PowerExchange Express CDC for Oracle parameters, use the sample configuration file, pwxorad.cfg, that PowerExchange supplies in the directory specified by the PWX_HOME environment variable or in the PowerExchange bin directory on the UNIX system.

Copy the sample file and customize the copy. If you copy the file under another name or to another directory, you must include the PARMFILE parameter in the ORAD CAPI_CONNECTION statement in the DBMOVER configuration file to point to the customized copy.

The following table summarizes the statements and options that you can use in this scenario:

Statement	Required or Optional	Description
DATABASE	Optional	Provides overrides for database connection information, including the capture connection string and database user ID and password. Also includes the TDE encryption password, which is required if you capture changes from TDE-encrypted tablespaces.
DICTIONARY	Required	Indicates where PowerExchange gets the Oracle data dictionary information that it uses to interpret the redo logs and how PowerExchange reacts to DDL changes that might occur after the dictionary is read into memory.
OPTIONS	Optional	Provides options for CDC processing, including the number of log records staged in memory, maximum spill file size, the period that the Express CDC waits for the next archived log to become available after the online redo log is overwritten, whether PowerExchange uses connection resiliency to retry killed Oracle sessions, whether Express CDC detects and removes invalid multibyte characters that appear at the end of a character field in an Oracle source table, whether PowerExchange captures Oracle rowid values and direct-path operations, and whether Express CDC ends or continues capture processing when encountering a DROP PARTITION operation.
READER	Required	Provides options for reading the redo logs.

In each statement that you include, you must specify at least one valid keyword. The end of an entire statement is indicated by a semicolon (;).

For complete information about all Express CDC statements and keywords, see the *PowerExchange CDC Guide for Linux, UNIX, and Windows*.

DATABASE Statement

The DATABASE statement specifies overrides for the capture connection string and the database user ID and password. The statement also provides options for capturing change data from TDE-encrypted tablespaces and from database instances in an Amazon Relational Database Service (RDS) for Oracle environment.

The DATABASE statement is optional. If you specify it, include at least one of its optional parameters.

If you want to capture change data from an Oracle Data Guard physical standby database that is open for read-only access, configure the DATABASE statement to connect to the standby database instance. If the standby database is *not* open for read-only access, configure the DATABASE statement to connect to the primary database instance and also define the STANDBY statement.

Syntax:

```
DATABASE
  [CONNECT_STRING=capture_connect_string]
  [EPWD=database_encrypted_password|PASSWORD=database_password]
  [RDS={N|Y}]
  [TDEWALLETDIR=TDE_wallet_path]
  [TDEWALLETEPWD=encrypted_password_for_TDE_wallet]
  [TDEWALLETPWD=cleartext_password_for_TDE_wallet]
  [USERID=database_user_id]
  ;
```

Parameters:

CONNECT_STRING

An Oracle connection string, defined in TNS, that the PowerExchange Express CDC for Oracle uses to connect to the Oracle database.

This connection string is overridden by the **Connect String** value, if specified, in the CDC session connection attributes.

EPWD

An encrypted password that PowerExchange uses to connect to the Oracle source database for PowerExchange Express CDC for Oracle. If you specify the USERID parameter in this statement, you must also specify either the EPWD or PASSWORD parameter.

This encrypted password overrides the **Password** value, if specified, in the CDC session connection attributes.

This encrypted password is overridden by the EPWD parameter value in the ORAD CAPI_CONNECTION statement, if specified.

PASSWORD

A clear text password that PowerExchange uses to connect to the Oracle source database for PowerExchange Express CDC for Oracle. If you specify the USERID parameter in this statement, you must also specify either the EPWD or PASSWORD parameter.

This password overrides the **Password** value, if specified, in the CDC session connection attributes.

This password is overridden by the PASSWORD parameter value in the ORAD CAPI_CONNECTION statement, if specified.

RDS

Indicates whether the Oracle source database is in an Amazon Service (RDS) for Oracle environment. Options are:

- **Y**. The source database instance is deployed in Amazon RDS for Oracle. PowerExchange Express CDC reads change data from the database redo logs that are in the ONLINELOG_DIR and ARCHIVELOG_DIR directories. You create these directories on the RDS file system.
- **N**. The source database instance is not deployed in Amazon RDS for Oracle.

Default is N.

TDEWALLETDIR

The fully qualified path and file name for the Oracle wallet file that is used for Oracle Transparent Data Encryption (TDE). Specify this parameter only if you capture change data from TDE-encrypted tablespaces

and either the Oracle wallet is not available to the database or the database is running on a server that is remote from Oracle redo logs from which Express CDC reads changes.

TDEWALLETEPWD

An encrypted password that PowerExchange requires to access the Oracle TDE wallet and get the master key that is required for reading and decrypting data from Oracle TDE-encrypted tablespaces. If you capture change data from TDE-encrypted tablespaces, you must specify either this parameter or the TDEWALLETPWD parameter. Do not specify both parameters.

Note: If you need to change the encryption password, first stop the PowerExchange Logger for Linux, UNIX, and Windows and the CDC session. Then edit the password, restart the PowerExchange Logger, and restart the CDC session.

TDEWALLETPWD

A clear text password that PowerExchange requires to access the Oracle TDE wallet and get the master key that is required for reading and decrypting data from Oracle TDE-encrypted tablespaces. If you capture change data from TDE-encrypted tablespaces, you must specify either this parameter or the TDEWALLETEPWD parameter. Do not specify both parameters.

Note: If you need to change this encryption password, first stop the PowerExchange Logger for Linux, UNIX, and Windows and the CDC session. Then edit the password, restart the PowerExchange Logger, and restart the CDC session.

USERID

A user ID that PowerExchange uses to connect to the Oracle source database for PowerExchange Express CDC for Oracle. If you specify the USERID parameter in this statement, you must also specify either the EPWD or PASSWORD parameter.

This user ID overrides the **User Name** value, if specified, in the CDC session connection attributes.

This user ID is overridden by the USERID parameter value in the ORAD CAPI_CONNECTION statement, if specified.

Usage Notes: You can specify the connection string in multiple locations. If you do so, PowerExchange uses the following order of precedence:

1. The **Connect String** value, if specified, in the CDC session connection attributes
2. The fourth positional parameter of the ORACLEID statement in the dbmover.cfg file
3. The CONNECT_STRING parameter value in the DATABASE statement of the PowerExchange Express CDC for Oracle configuration file
4. The ORACLE_SID environment variable

If you specify none of these values, PowerExchange passes Null values in the OCI call.

DICTIONARY Statement

The DICTIONARY statement indicates where PowerExchange Express CDC for Oracle gets the data dictionary information that it uses to interpret redo logs. This statement also controls how PowerExchange Express CDC for Oracle reacts when it encounters DDL changes in the redo log records.

The DICTIONARY statement is required.

Syntax:

```
DICTIONARY
  MODE=STATIC
  SOURCE=ONLINE
  [EXCEPTIONS={FAIL|WARN}]
  ;
```

Parameters:

MODE

Required. Indicates whether PowerExchange Express CDC for Oracle expects the data dictionary to remain the same or change after reading it into memory. The only valid value is STATIC. In STATIC mode, PowerExchange expects the data dictionary to remain the same. If structural changes to source tables occur, the EXCEPTIONS parameter determines whether PowerExchange fails or issues a warning. However, PowerExchange Express CDC always tolerates the following DDL changes and continues CDC processing, regardless of the EXCEPTIONS setting:

- ALTER TABLE ADD statements for adding one or more columns to a table
- ALTER TABLE ADD PARTITION statements
- ALTER TABLE ADD CONSTRAINT statements
- CREATE USER statements
- ALTER USER statements
- DROP USER statements

SOURCE

Required. Indicates where PowerExchange Express CDC for Oracle gets its data dictionary information. The only valid value is ONLINE, which indicates that PowerExchange Express CDC for Oracle gets the data dictionary information from the Oracle online system when the change capture process initializes.

EXCEPTIONS

Optional. When MODE=STATIC, controls whether PowerExchange Express CDC for Oracle fails or continues with a warning message when it detects that a structural change was made to an Oracle table from which change data is captured.

This parameter does not apply to ADD PARTITION changes. PowerExchange Express CDC for Oracle tolerates ADD PARTITION changes.

Valid values are:

- **FAIL**. CDC processing ends abnormally.
- **WARN**. PowerExchange prints a warning message, and CDC processing continues. This option can result in loss of change data. Use this option only at the direction of Informatica Global Customer Support.

Default is FAIL.

OPTIONS Statement

The OPTIONS statement specifies parameters for controlling CDC processing. The parameters control memory use, spill file size, capture of rowid values, long outstanding UOWs, and the wait period for advancing restart tokens when no change capture activity is occurring.

The OPTIONS statement is optional, and all of the parameters in the statement are optional. If you specify the OPTIONS statement, include at least one parameter.

Syntax:

```
OPTIONS
  [AGEOUTPERIOD=minutes]
  [CONNRETRYMAX=number]
  [CONNRETRYWAIT=seconds]
  [LARGEOPS=number_of_operations]
  [LOGARCHIVEWAIT=seconds]
  [MEMOPS=number_of_log_records]
  [MONITOR_INTERVAL=minutes]
```

```
[PARTITION_DROP_FAIL={Y|N}]
[REPORTDDL={Y|N}]
[RETRYONKILLSESSION={Y|N}]
[ROWID={Y|N}]
[ROW_MOVEMENT_FAIL={Y|N}]
[RSTRADV=seconds]
[SPILL_FILE_PREFIX=spill_file_name_prefix]
[SPILLMAX=kilobytes]
[SUPPORT_DIRECT_PATH_OPS={Y|N}]
[TIME_STAMP_MODE={LOGTIME|COMMITTIME|BEGINTIME}]
[TRUNCINVALIDCHARS={Y|N}]
;
```

Parameters:

AGEOUTPERIOD=*minutes*

The age, in number of minutes, that an outstanding UOW that has no change records of CDC interest must reach before it is removed from the calculation of the CDC restart point. The age is calculated as the time difference between the start of the outstanding UOW and the start of the most recent UOW. This age-out processing occurs during the monitoring interval.

Use this parameter to prevent CDC failures that can occur if you shut down and then restart capture processing while the transaction is outstanding. After the restart, the archived redo log in which the outstanding UOW started might not be available, causing the Express CDC log reader to fail.

Valid values are 60 to 43200. By default, no value is specified and this parameter is disabled.

Note: Oracle stores all time values in the log in local time. As a result, at the beginning or end of daylight saving time, a UOW might age out an hour late or an hour early.

CONNRETRYMAX

The maximum number of times that PowerExchange tries to reconnect to an Oracle source database or ASM instance. Use this parameter in conjunction with the CONNRETRYWAIT parameter if you get multiple PWX-36086 messages followed by an Oracle error that indicates a dropped connection or if you set the RETRYONKILLSESSION parameter to Y.

Note: The CONNRETRYMAX, CONNRETRYWAIT, and RETRYONKILLSESSION parameters help improve connection resiliency.

Valid values are 0 to 86400. A value of 0 results in no retries. Default is 12.

CONNRETRYWAIT

The number seconds that PowerExchange waits between attempts to reconnect to an Oracle source database or ASM instance. Use this parameter in conjunction with CONNRETRYMAX parameter if you get multiple PWX-36086 messages followed by an Oracle error that indicates a dropped connection or if you want to tune connection retries for killed sessions. This parameter can help improve connection resiliency. Valid values are 1 to 300. Default is 5 seconds.

LARGEOPS

Overrides the default value that PowerExchange uses to identify transactions as large transactions for reporting purposes. Enter the number of log records that a transaction must process to be considered a large transaction. A log record might contain multiple DML operations or part of a single operation.

PowerExchange issues status messages for large transactions that meet this criteria. If PowerExchange issues too many messages, you can increase this value to limit the number of messages.

Valid values are 1 through 2147483 (1000 through 2,147,483,000 operations). The default value is one fifth of the MEMOPS value, rounded to the nearest thousand. For example, if the MEMOPS value is 5120, the default LARGEOPS value is 1000 (1,000,000 operations).

LOGARCHIVEWAIT

After an Oracle online redo log starts being overwritten, the number of seconds that PowerExchange Express CDC waits for the copy of the log to become available as a new archived redo log for change data capture processing. In an Oracle Data Guard environment, if Express CDC captures change data from a physical standby database, this parameter specifies the number of seconds that Express CDC waits for the next archived redo log to be transported from the primary database to the standby database.

Valid values are 0 through 86400. Default is 30. If you use any value less than the value of the STATUSCHECKINTERVAL parameter in the READER statement, Express CDC waits for the STATUSCHECKINTERVAL period.

MEMOPS

The maximum number of redo log records that contain DML operations that PowerExchange can stage in memory while processing Oracle transactions.

Valid values are 1000 through 1048576. Default is 5120.

MONITOR_INTERVAL

The time interval, in minutes, at which PowerExchange checks transaction activity for long outstanding transactions and large transactions. A long outstanding transaction is one that remains active for two monitoring intervals, and a large transaction is one that meets the LARGEOPS criteria. When this interval elapses, PowerExchange issues messages that identify the large transactions and long outstanding transactions and report their processing activity. PowerExchange also issues messages that identify the current position in the change stream. Valid values are 0 through 720. A value of 0 disables monitoring. Default is 5.

PARTITION_DROP_FAIL

Controls whether PowerExchange Express CDC for Oracle ends with an error or continues processing when the log reader encounters an ALTER TABLE DROP PARTITION operation for a registered Oracle source table.

Options:

- **Y**. Express CDC processing ends with error message PWX-36332, which reports the log position of the DROP PARTITION operation that caused the failure.
- **N**. Express CDC ignores the DROP PARTITION operation and continues change capture processing. Message PWX-36390 reports that Express CDC encountered a DROP PARTITION operation for a source table.

Default is Y.

REPORTDDL={Y|N}

Controls whether PowerExchange Express CDC reports all of the DDL operations that it encounters in the Oracle redo logs for Oracle source tables with active capture registrations. Express CDC writes the following information for each DDL operation to a file that is generated in the directory from which Express CDC runs: the DDL statement, log position, owner number, DDL object number, and sequence number. The file naming conventions are:

- For RAC systems:

  `PWX_ORL_DDL_Dyyyymmdd_Thhmmss.MBRnode_sequence#.rpt`

- For non-RAC systems:

  `PWX_ORL_DDL_Dyyyymmdd_Thhmmss.sequence#.rpt`

In these file names, the *sequence#* is a generated number that starts from 0001 and that is incremented by 1 for each new file. A new file is generated every 20 MB of DDL change records.

Options are:

- **Y**. Generate the report of DDL operations for registered source tables.
- **N**. Do not generate the report of DDL operations.

Default is N.

RETRYONKILLSESSION

Controls whether PowerExchange Express CDC can detect when an Oracle KILL SESSION event occurs for a specific PowerExchange connection to an Oracle source instance and then retry the connection so that the Express CDC log reader and PowerExchange Logger for Linux, UNIX, and Windows process do not end abnormally. A KILL SESSION event occurs when a user issues the following SQL statement:

```
ALTER SYSTEM KILL SESSION 'sid,serial_number' [IMMEDIATE]
```

In this statement, the variable *sid* is the session ID and the variable *serial_number* is the session serial number, as shown in the V$SESSION view.

Options:

- **N**. PowerExchange does not try to re-establish killed sessions for Express CDC processes. If a KILL SESSION event occurs, the PowerExchange Express CDC log reader and PowerExchange Logger end abnormally.
- **Y**. PowerExchange tries to re-establish killed sessions for Express CDC processes. Also set the CONNRETRYMAX parameter to a value greater than 0 to indicate the maximum number of times that PowerExchange retries the connection to the source instance.

  Tip: The RETRYONKILLSESSION, CONNRETRYMAX, and CONNRETRYWAIT parameters help improve connection resiliency.

Default is N.

Important: Before setting this parameter to Y, consult with your Oracle database administrator to ensure that no unintended consequences occur.

ROWID

Controls whether Oracle physical rowid values are included in captured change records for tables that do not have Oracle row movement enabled. PowerExchange writes the rowid values to the PowerExchange-generated DTL__CAPXROWID column.

For example, you might want to use this parameter if you have unkeyed source tables on which you need to perform some processing that requires a unique row ID when extraction sessions run.

Valid values are:

- **N**. Do not capture rowid values. The DTL__CAPXROWID column contains null values.
- **Y**. Capture rowid values. The DTL__CAPXROWID column contains rowid values for tables that do not have row movement enabled.

Default is N.

ROW_MOVEMENT_FAIL

When ROWID=Y, controls whether PowerExchange Express CDC for Oracle processing fails or continues if PowerExchange detects that row movement is enabled for a source table.

You might want to continue processing if you do not need to capture rowid values for tables that have row movement enabled.

Valid values are:

- **Y**. CDC processing fails if PowerExchange encounters a table with row movement enabled.
- **N**. CDC processing continues. For any table that has row movement enabled, the DTL__CAPXROWID column contains null values.

Default is Y.

RSTRADV

The number of seconds that PowerExchange waits before advancing restart and sequence tokens for a registered data source during periods when UOWs do not include any changes of interest for the data source. When the wait interval expires, PowerExchange returns the next committed "empty UOW," which includes only updated restart information.

PowerExchange resets the wait interval to 0 when one of the following events occur:

- PowerExchange completes processing a UOW that includes changes of interest.
- PowerExchange returns an empty UOW because the wait interval expired without PowerExchange receiving any changes of interest.

Valid values are 0 through 86400. No default is provided.

If you do not specify RSTRADV, PowerExchange does not advance restart and sequence tokens for a registered source during periods when PowerExchange receives no changes of interest. In this case, when PowerExchange warm starts, it reads all changes, including those not of interest for CDC, from the restart point.

Note: A value of 0 can degrade performance. In addition to the UOWs that contain changes for registered sources of interest, PowerExchange returns an empty UOW for every UOW that does not contain changes for the registered sources of interest.

SPILL_FILE_PREFIX

A file name prefix to a temporary file path for the spill files. If you want to specify the directory where the spill files reside, specify the fully qualified path to the temporary directory.

Linux or UNIX example:

```
SPILL_FILE_PREFIX=”/tmp/spillfiles/”
```

Windows example:

```
SPILL_FILE_PREFIX=”C:\tmp\spillfiles\xxx”
```

Default path is the TMP or TMPDIR environment variable value. If TMP or TMPDIR is not present, the configuration uses the current directory.

SPILLMAX

The maximum size, in kilobytes, of a single PowerExchange Express CDC for Oracle spill file on UNIX. PowerExchange writes transactions to spill files when it does not have sufficient memory to store them. After PowerExchange processes all of the transactions in a spill file, the spill file is freed.

A large spill file retains disk space longer before being freed than a smaller-sized spill file.

Valid values are 1 through 2097151. Default is 10240.

SUPPORT_DIRECT_PATH_OPS

Controls whether Express CDC captures or ignores Oracle direct-path operations, such as direct-path Inserts, on registered source tables.

Options:

- **N**. Disables the capture of direct-path operations. If the log reader encounters a redo log record for a direct-path operation, Express CDC ignores the direct-path operation with an informational message and continues.
- **Y.** Enables the capture of direct-path operations.

  Note: Express CDC does not capture direct-path operations for tables that use Oracle Exadata Hybrid Columnar Compression (EHCC).

Default is N.

TIME_STAMP_MODE

The type of timestamp that PowerExchange records in the generated DTL__CAPXTIMESTAMP column of each change record for a transaction. Usually, you specify this parameter only if you want to display the Oracle commit timestamp instead of the Oracle log timestamp.

Options are:

- **LOGTIME**. The timestamp that Oracle periodically writes to the Oracle archived logs.
- **COMMITTIME**. The timestamp of the transaction commit on the source database. Specify this option if you use the timestamp to calculate latency.
- **BEGINTIME**. The timestamp of the begin UOW log record.

Default is LOGTIME.

TRUNCINVALIDCHARS

Indicates whether to detect and remove invalid multibyte characters that appear at the end of a character field in an Oracle source table. The characters are invalid because they have been truncated. If you allow these invalid characters to be passed to a PowerCenter workflow that has an Oracle target, PowerCenter might corrupt subsequent columns in the target table when running in Unicode mode.

Valid values are:

- **Y**. Remove invalid multibyte characters from the source data. The invalid characters are not passed to PowerCenter or applied to the target database.
- **N**. Capture the invalid multibyte characters and pass them to PowerCenter. When the PowerCenter workflow tries to write the invalid characters to the target, PowerCenter might corrupt data in the subsequent columns in the target table.

Default is N.

READER Statement

The READER statement provides parameters that the PowerExchange Express CDC log reader uses to read redo log files, issue log-read status messages, connect to an Oracle Automatic Storage Management (ASM) instance, and check the status of the Oracle server during periods of inactivity.

This topic describes the parameters that are used in this scenario. For descriptions of all PowerExchange READER statement parameters, see the *PowerExchange CDC Guide for Linux, UNIX, and Windows*.

The READER statement is required.

Syntax:

```
READER
   MODE={ACTIVE}
  ;
```

Parameters:

MODE

Required. An option that indicates the source of and types of redo logs that the PowerExchange Express CDC log reader reads.

- **ACTIVE**. The PowerExchange Express CDC for Oracle log reader reads active and archived redo logs from the Oracle online system. With this option, you can run the PowerExchange for Linux, UNIX, and Windows in either continuous or batch mode. Optionally, you can use the ACTIVELOGMASK parameter to filter the active redo logs and use the ARCHIVEDEST parameter to limit the archived log destinations from which to read archived logs. For descriptions of the ACTIVELOGMASK and ARCHIVEDEST parameters, see the *PowerExchange CDC Guide for Linux, UNIX, and Windows*.

  Default is ACTIVE.

Example PowerExchange Express CDC for Oracle Configuration File

This example PowerExchange Express CDC for Oracle configuration file contains the required statements only.

```
DICTIONARY
  MODE=STATIC
  SOURCE=ONLINE;
READER
  MODE=ACTIVE;
```

Customizing the PowerExchange Logger Configuration File

To specify PowerExchange Logger parameters, use the sample configuration file, pwxccl.cfg, that PowerExchange supplies in the directory specified by the PWX_HOME environment variable on the UNIX system.

Copy the sample file under another name and then customize the copy. To identify a custom configuration file name and path for PowerExchange Logger use, specify the CS parameter when you start the Logger.

Logger Parameter Descriptions

This topic describes only the parameters that are used in this scenario.

For descriptions of all PowerExchange Logger parameters, see the *PowerExchange CDC Guide for Linux, UNIX, and Windows*.

The parameter descriptions use the following conventions:

- *Italics* indicate a variable.
- Curly braces { } enclose alternative options. Enter only one option. Do not type the braces when you enter the option.
- A vertical bar | indicates a mutually exclusive choice. When used with braces, you must enter one of the items.
- Underlining indicates a default value.

Note: If a parameter value contains one or more spaces, you must be enclose it in double quotation marks (").

CAPT_IMAGE={AI|BA}

Type of data images that the PowerExchange Logger stores in its log files. Use this parameter to control whether the PowerExchange Logger stores after images only or both before and after images of the data in its log files.

This parameter affects the amount of storage that you use for PowerExchange Logger log files and whether before image data is available for use in extraction processing.

Enter one of the following options:

- **AI**. Stores only after images in the PowerExchange Logger log files.
- **BA**. Stores both before and after images in the PowerExchange Logger log files.

Default is **AI**. With **AI**, less storage is required for PowerExchange Logger log files. However, the following CDC limitations apply:

- You cannot use before images of the data in extraction processing. If you add before image (BI) fields to extraction maps, PowerCenter CDC sessions that reference the BI fields fail.
- If you add change indicator (CI) fields to extraction maps, PowerCenter CDC sessions that reference the CI fields fail.

Informatica recommends that you enter **BA** if you have sufficient storage for larger log files. The sample pwxccl configuration file that PowerExchange supplies specifies **BA**.

Note: If you use **BA** and add CI columns to the extraction maps, any Insert and Delete operations on the source result in Null values in the CI columns. Any Update operations on the source result in the Y or N indicator in the CI columns.

CAPTURE_NODE_EPWD=*encrypted_password*

An encrypted password that is associated with the user ID specified in the CAPTURE_NODE_UID parameter. This password, in conjunction with the CAPTURE_NODE_UID value, is used to control PowerExchange access to capture registrations and change data.

Tip: You can create an encrypted password in the PowerExchange Navigator by selecting **File** > **Encrypt Password**.

Note: If you specify CAPTURE_NODE_EPWD, do not also specify CAPTURE_NODE_PWD.

CAPTURE_NODE_PWD=*password*

A clear text password that is associated with the user ID specified in the CAPTURE_NODE_UID parameter. This password, in conjunction with the CAPTURE_NODE_UID value, is used to control PowerExchange access to capture registrations and change data.

Note: If you specify CAPTURE_NODE_PWD, do not also specify CAPTURE_NODE_EPWD.

CAPTURE_NODE_UID=*user_id*

User ID that is used to control access to capture registrations and change data on the local machine.

For a UNIX node with an Oracle source, enter the ORACAPTL1 user ID that you defined, which permits access to the Oracle online and archive redo logs.

Note: If you specify CAPTURE_NODE_UID, you must enter a password or encrypted password in either the CAPTURE_NODE_PWD or CAPTURE_NODE_EPWD parameter, but not both.

COLL_END_LOG={0|1}

Required. The PowerExchange Logger operational mode.

Enter one of the following options:

- **0**. Runs the PowerExchange Logger continuously until you manually stop it. After the Writer subtask completes a processing cycle, it waits for the number of minutes specified in the NO_DATA_WAIT parameter before starting another processing cycle.
- **1**. Runs the PowerExchange Logger in batch mode. The PowerExchange Logger shuts down after the seconds specified in the NO_DATA_WAIT2 parameter elapse and no data has been received.

Default is **0** for continuous mode.

COND_CDCT_RET_P={*days*|60}

Recommended. Retention period, in days, for CDCT records and PowerExchange Logger log files. Log files that are older than this period and their corresponding CDCT records are deleted automatically during PowerExchange Logger cleanup processing. Cleanup processing occurs during startup, file switch, or shutdown processing.

Enter a number greater than 0. Default is 60.

When setting this parameter, try to minimize the size of the CDCT file while preserving the log files that contain the earliest change data that you might need to access. Use the following guidelines:

- If you set the retention period to a low value, ensure that PowerExchange extracts change data from the PowerExchange Logger log files during this period. Otherwise, the log files for which the retention period has elapsed are deleted and you can lose change data. For example, if the retention period is 5 and you plan not to run extractions during a 10-day holiday, increase the retention period to 15. This approach ensures that the log files with the change data you need are not deleted until the extractions run again.
- If you set the retention period to a high value, the CDCT can become very large, depending on the number active capture registrations. Also, the number of PowerExchange Logger log files might increase. For continuous extraction mode, you can use the PowerExchange Logger FILE_SWITCH parameters to decrease the number of log files and increase their size instead.
- If you use continuous extraction mode, PowerExchange reads the CDCT file each time the interval specified in the FILEWAIT parameter of the CAPX CAPI_CONNECTION statement elapses. If a CDCT file becomes large, this read activity can increase I/O, system resource use, and latency of change data extraction. If you use batch extraction mode, this high read activity is not a consideration.

CONDENSENAME=*service_name*

Optional. A name for the command-handling service for a PowerExchange Logger for Linux, UNIX, and Windows process to which infacmd pwx or pwxcmd commands are issued.

Enter a service name up to 64 characters in length. No default is available.

This service name must match the service name that is specified in the associated SVCNODE statement in the dbmover configuration file. The SVCNODE statement specifies the TCP/IP port on which this service listens for infacmd pwx or pwxcmd commands.

Tip: If you run the PowerExchange Logger as a background process in continuous mode, specify this parameter so that you can use the pwxcmd program to issue commands to the PowerExchange Logger. Without the use of pwxcmd, you cannot shut down a PowerExchange Logger process that is running in the background or send status information to a computer that is remote from where the PowerExchange Logger runs.

CONN_OVR=*capi_connection_name*

Recommended. The name of the override CAPI_CONNECTION statement to use for the PowerExchange Logger. If you do not enter CONN_OVR, the PowerExchange Logger uses the default CAPI_CONNECTION in the dbmover configuration file, if specified.

Enter a valid CAPI_CONNECTION name for the source type.

Informatica recommends that you specify CONN_OVR because it is the only type of override that the PowerExchange Logger can use.

DBID=*instance_name*

Required. A source identifier, sometimes called the *instance* name, that is defined in capture registrations. When used with DB_TYPE, it defines selection criteria for capture registrations in the CCT file.

For an Oracle source on UNIX, enter the **Instance** name that is displayed for the registration group in the **Resource Inspector**. Ensure that this value matches the first positional parameter in the ORACLEID statement in the dbmover configuration file.

DB_TYPE={ADA|AS4|DB2|DCM|IDL|IMS|MSS|ORA|UDB|VSM}

Required. The source database type. For an Oracle source, enter ORA.

EXT_CAPT_MASK=*path/prefix*

Required. An existing directory path and a unique prefix to be used for generating the PowerExchange Logger log files.

Maximum length is 256 characters.

For example:

```
/capture/pwxlog
```

Note: If you enter a value that include spaces, you must enclose the value in double quotation marks (").

Verify that no existing files match this path and prefix. PowerExchange considers any file that matches this path and prefix to be a PowerExchange Logger log file, even if it is unrelated to PowerExchange Logger processing.

To create the log files, the PowerExchange Logger appends the following information:

```
.CND.CPyymmdd.Thhmmssnnn
```

Where:

- *yymmdd* is a date composed of a two-digit year, a month, and a day.
- *hhmmss* is 24-hour time value, including hours, minutes, seconds.
- *nnn* is a generated sequence number that ensures uniqueness of the file name.

For example:

```
/capture/pwxlog.CND.CP080718.T1545001
```

FILE_FLUSH_VAL={*seconds*|-1}

Recommended. File flush interval in seconds. When this interval elapses, the PowerExchange Logger writes any outstanding change data that it read from the source to log files on disk. After the change data is flushed to disk, CDC sessions that use continuous extraction mode can read the change data. This parameter affects the latency of continuous change data extractions.

Valid values are:

- -1. The PowerExchange Logger does *not* flush outstanding change data to the current log file on disk based on this parameter. Enter this value only if you use batch extraction mode. If you use continuous extraction mode, this value can increase the latency of your continuous extraction sessions.
- 0. The PowerExchange Logger flushes outstanding change data after every record.
- Any value from 1 through 86400. The PowerExchange Logger flushes outstanding change data at the specified interval.

Default is -1.

Warning: A value of 0 can degrade the performance of the PowerExchange Logger and file system.

Set this value as appropriate for your CDC environment. Values that are too high can increase change extraction latency, and values that are too low can degrade PowerExchange Logger and system performance. Informatica recommends that you set this parameter to a value that is equal to or greater than the NO_DATA_WAIT2 value because file flushes cannot occur until the NO_DATA_WAIT2 period expires.

FILE_SWITCH_CRIT={M|R}

Type of units to use for the FILE_SWITCH_VAL parameter, which determines when to do an automatic file switch.

Enter one of the following options:

- **M** for minutes.
- **R** for records.

Default is **M**.

FILE_SWITCH_VAL={*minutes_or_records*|30}

Number of minutes or change records, as determined by FILE_SWITCH_CRIT, that must elapse before PowerExchange performs a file switch.

Enter a number greater than 0. Default is 30.

For example, if you use 30 and enter FILE_SWITCH_CRIT=R, the PowerExchange Logger performs a file switch every 30 records. If FILE_SWITCH_CRIT=M, the PowerExchange Logger performs a file switch every 30 minutes.

If the PowerExchange Logger log files contain no data when the FILE_SWITCH_VAL threshold is reached, the file switch does not occur.

This value affects the size of the PowerExchange Logger log files. Specify a value that results in log files of the appropriate size for your environment.

Tip: When using continuous extraction mode, set this parameter such that you have larger log files and a smaller CDCT file. When using batch extraction mode, set this parameter to a value that causes file switches to occur within the time frame that meets your change extraction latency requirements.

LOGGER_DELETES_EXPIRED_CDCT_RECORDS={Y|N}

Controls whether the PowerExchange Logger deletes CDCT records for log files for which the retention period has expired.

Enter one of the following options:

- **Y**. The PowerExchange Logger deletes expired CDCT records during file switches. You cannot use the PWXUCDCT utility DELETE_EXPIRED_FILES command to manually delete expired log files and their related CDCT records.
- **N**. The PowerExchange Logger does not delete expired CDCT records. However, you can use the PWXUCDCT utility DELETE_EXPIRED_FILES command to manually delete expired log files and their related CDCT records.

  Note: This parameter does not affect PowerExchange Logger deletions of CDCT records that are rolled back because of a cold start or a warm start to a prior point in time.

Default is **Y**.

NO_DATA_WAIT={*minutes*|1}

If you run the PowerExchange Logger in continuous mode, specify the number of minutes that the PowerExchange Logger must wait before starting the next logging cycle.

Enter 0 or a number greater than 0. Default is 1.

A value of 0 causes no waiting to occur between PowerExchange Logger processing cycles. If source data is not available, the CAPI sleeps.

For continuous extraction mode, enter 0 for no waiting, or enter a low value so that the next logging cycle starts shortly after the current one completes.

If the value of FILE_SWITCH_CRIT is M and the value of FILE_SWITCH_VAL is less than the value of NO_DATA_WAIT, the PowerExchange Logger uses the FILE_SWITCH_VAL value instead.

NO_DATA_WAIT2={*seconds*|600}

The number of seconds that PowerExchange waits at the end-of-log for more change data before returning control to the PowerExchange Logger. If this wait period elapses and no new change data has been received, PowerExchange returns control to the PowerExchange Logger, and the PowerExchange Logger then stops the current logging cycle.

Enter a number greater than 0. Default is 600.

Informatica recommends a value of 10. If you enter a greater value, execution of commands for the PowerExchange Logger might be delayed.

Use the same value for the FILE_FLUSH_VAL parameter.

PROMPT={Y|N}

When you run the PowerExchange Logger in foreground mode, controls whether PowerExchange displays a user confirmation prompt and waits for a response when you perform one of the following actions:

- Cold start the PowerExchange Logger.
- Warm start the PowerExchange Logger from a previous position in the change stream. This situation occurs only if the CDCT file still contains records related to the deleted files.

Enter one of the following options:

- **Y**. Displays the confirmation message PWX-33236 for a cold start or PWX-33242 for a warm start. You must respond to the message for startup processing to continue.
- **N**. Does not display the confirmation messages. PowerExchange attempts to start without first prompting for user confirmation.

If you run the PowerExchange Logger in foreground mode, the default is Y.

If you run the PowerExchange Logger in background mode or as a PowerExchange Logger Service in the Informatica domain, the default is N. In this case, if you enter PROMPT=Y in the pwxccl.cfg file, the PowerExchange Logger ignores this setting, issues error message PWX-33253, and continues processing.

RESTART_TOKEN and SEQUENCE_TOKEN

A pair of token values that define a restart point for starting change data processing when a PowerExchange Logger is cold started.

Depending on how you set these parameters, PowerExchange Logger processing starts from one of the following restart points during a cold start:

- If you do not specify these parameters, processing starts from the current end-of-log position.
- If you enter 0 for both parameters, processing starts from the default start location. If you use Express CDC to capture data from an Oracle source, the default location is the beginning of the most recent archive log.
- If you enter restart token and sequence token values other than 0, processing resumes from the specific restart point defined by these token values. To perform a special start of the PowerExchange Logger, you must specify the RESTART_TOKEN and SEQUENCE_TOKEN parameters with specific values, and the SEQUENCE_TOKEN value must be greater than the sequence token in the CDCT file.

STATS=(MONITOR[,*interval*|0])

Enables PowerExchange Logger collection of the following monitoring statistics:

- PowerExchange Logger process ID (PID)

- Status of the PowerExchange Logger Writer task
- CPU time used by the PowerExchange Logger - total and for Writer task functions
- Memory use (current/total/maximum) in kilobtyes, total and for the Controller, Command Handler, and Writer tasks
- Record counts, including the number of inserts, updates, deletes, and commits that the PowerExchange Logger processed - total since the Logger started and for the current active log file and the active logging cycle

Also enables collection of the following statistics for PowerExchange Logger group definitions, if defined:

- The number of DML operations and commits processed for each group
- Then number of change records that have not yet been flushed to a Logger log file on disk
- The name of the open Logger log file for each group and the file open timestamp

The statistics are printed to the PowerExchange message log and on screen when you enter the DL or DG command at the command line, or enter the pwxcmd displaystats -tp {logger|groups} command from a remote Linux, UNIX, or Windows system.

For more information about the monitoring commands and related reports, see the *PowerExchange Command Reference*.

Optionally, you can include the *interval* subparameter to publish the Logger statistics at a regular interval.

{*interval*|0}

Optional. The interval, in minutes, after which PowerExchange publishes monitoring statistics for the PowerExchange Logger. The interval-based statistics that are written to the PowerExchange message log file are the same as those published by the DL (or DS) command and pwxcmd displaystats -tp logger command. However, a subset of the message output is displayed on screen to prevent flooding the screen with messages over time.

Valid values are 0 through 120. Default is 0, which disables interval-based reporting of PowerExchange Logger monitoring statistics. With the default value, PowerExchange writes these statistics only when a DL or pwxcmd displaystats -tp logger command is issued.

Example PowerExchange Logger Configuration File

The following example PowerExchange Logger configuration file shows statements for this scenario:

```
DBID=ORACOL01
DB_TYPE=ORA

CAPTURE_NODE_UID=ORACAPTL1
CAPTURE_NODE_PWD=Oracaptl1

PROMPT=Y

/* Required to issue pwxcmd commands to the Logger
CONDENSENAME=pwxccl_svc

/* The directory that you created for the Logger log files.
EXT_CAPT_MASK=/capture/pwxccl_log
/* How long to retain CDCT entries and CND files.
COND_CDCT_RET_P=60
LOGGER_DELETES_EXPIRED_CDCT_RECORDS=Y

CONN_OVR=CAPORA3

/* 0 = continuous, 1 = Stop at end-of-log (batch)
COLL_END_LOG=0
```

```
/* Number of minutes that the Logger in continuous mode waits between CAPI read cycles
NO_DATA_WAIT=0
/* Number of seconds that the Logger waits at the end-of-log for more change data
NO_DATA_WAIT2=4

/* Number of seconds that the Logger waits before flushing data to the current log file on disk
/* -1 = No flush, 0 = flush every record, 1 to 86400 = flush every N seconds
FILE_FLUSH_VAL=5
FILE_SWITCH_CRIT=M
FILE_SWITCH_VAL=20

CAPT_IMAGE=BA

/* Include the following parameter to collect Logger monitoring statistics and publish them
/* to the PowerExchange message log every 60 minutes.
/* STATS=(MONITOR,60)

/* Enter 00s to use the beginning of the most recent Oracle archive log as the restart location
/* SEQUENCE_TOKEN=00
/* RESTART_TOKEN=00
```

Customizing the DBMOVER Configuration File on UNIX

To configure the DBMOVER statements and parameters for the PowerExchange Listener, use the sample DBMOVER configuration file in the PowerExchange installation directory on the UNIX machine.

Copy the sample file and customize the copy. If you copy the file under another name or to another directory, configure the CONFIG parameter to point to your customized file when you start the PowerExchange Listener.

For this scenario, the following DBMOVER statements are required:

- CAPT_PATH
- CAPT_XTRA
- CAPX CAPI_CONNECTION
- LISTENER
- ORACLE_CAPTURE_TYPE
- ORACLEID
- ORAD CAPI_CONNECTION

The following additional DBMOVER statements are recommended:

- CAPI_CONN_NAME
- CMDNODE
- LOGPATH
- SVCNODE
- TRACING

This article describes only these required and recommended statements. For information about all DBMOVER statements, see the *PowerExchange Reference Manual*.

DBMOVER Statement Descriptions

The following statements are required or recommended in the DBMOVER configuration file on the UNIX system:

CAPI_CONNECTION - CAPX

Specifies a named set of parameters that the Consumer API (CAPI) uses for continuous extraction of change data from PowerExchange Logger for Linux, UNIX, and Windows log files. Required for continuous extraction mode.

Syntax:

```
CAPI_CONNECTION=([DLLTRACE=trace_id]
                 ,NAME=capi_connection_name
                [,TRACE=trace_name]
                 ,TYPE=(CAPX
                       ,DFLTINST=instance_name
                       [,FILEWAIT={seconds|1}]
                       [,NOSEQVAL={N|Y}]
                       [,RSTRADV=seconds]
                       [,RSTRANMODE={N|Y}]
                 )
)
```

Parameters for this scenario:

- NAME. A unique user-defined name for the CAPI_CONNECTION statement. Maximum length is eight alphanumeric characters.
- TYPE. The type of CAPI_CONNECTION statement. For continuous extraction mode with the PowerExchange Logger, this value must be CAPX.
- DFLTINST. The Oracle source instance identifier that is specified in the **Instance** field for the registration group. This value must match the DBID value that is specified in the PowerExchange Logger configuration file. Maximum length is eight characters.

CAPI_CONNECTION - ORAD

Specifies a named set of parameters that the Consumer API (CAPI) uses to connect to the change stream and control PowerExchange Express CDC for Oracle processing for Oracle data sources.

Syntax:

```
CAPI_CONNECTION=([DLLTRACE=trace_id]
                 ,NAME=capi_connection_name
                 ,TYPE=(ORAD
                       [,EPWD=database_encrypted_password]
                        ,ORACOLL=collection_id
                       [,PARMFILE=express_cdc_configuration_file]
                       [,PASSWORD=database_password]
                       [,USERID=database_user_id]
                  )
)
```

Parameters:

- NAME. A unique user-defined name for this CAPI_CONNECTION statement. Maximum length is eight alphanumeric characters.
- TYPE. Required. The type of CAPI_CONNECTION statement. For PowerExchange Express CDC for Oracle sources, this value must be ORAD.
  - ORACOLL. The collection identifier for the Oracle instance. This value must match the collection ID in the first positional parameter of an ORACLEID statement in the same DBMOVER configuration file.
  - PARMFILE. The path and file name for the PowerExchange Express CDC for Oracle configuration file, relative to the current working directory. You can use this parameter to override the default path and file name or to remind PowerExchange users of the default path and file name. If this parameter is not specified, PowerExchange uses *pwx_installation*\pwxorad.cfg by default. The default path is the path in the PWX_HOME environment variable, or if this environment variable is not defined, the default path is the path to the PowerExchange bin directory.
  - PASSWORD or EPWD. Optional. A clear text password or encrypted password (EPWD) that PowerExchange uses to connect to the Oracle source database for PowerExchange Express CDC for Oracle. If you specify the USERID parameter in this statement, you must specify either the EPWD or PASSWORD parameter.

The encrypted password overrides the EPWD parameter value in the DATABASE statement of the PowerExchange Express CDC for Oracle configuration file and the **Password** value in the CDC session connection attributes.

- USERID. A user ID that PowerExchange uses to connect to the Oracle source database for PowerExchange Express CDC for Oracle. If you specify the USERID parameter in this statement, you must also specify either the EPWD or PASSWORD parameter.

Note: You cannot define both ORAD CAPI_CONNECTION and ORCL CAPI_CONNECTION statements in the same DBMOVER configuration file.

CAPI_CONN_NAME

Specifies the name of the CAPI_CONNECTION statement that PowerExchange uses by default for all data source types when you do not supply a CAPI connection override.

Syntax:

```
CAPI_CONN_NAME=capi_connection_name
```

For the *capi_connection_name* variable, enter the name of the CAPI_CONNECTION statement that PowerExchange uses when you do not specify a CAPI_CONNECTION override value. In this scenario, enter the name of the ORAD CAPI_CONNECTION statement.

CAPT_PATH

Specifies the path to a local directory on the UNIX system where you want to store CDC control files.

```
CAPT_PATH=path/directory
```

For the *path* variable, enter the path and directory name for storing the following control files:

- CCT file, which contains capture registrations.
- CDEP file, which contains application names for any PowerCenter extractions that use ODBC connections.
- CDCT file, which contains information about PowerExchange Logger for Linux, UNIX, and Windows log files and restart information.

Default is the PowerExchange installation directory.

Tip: Informatica recommends that you use a unique directory name to separate these CDC objects from the PowerExchange code. This practice makes migrating to a another PowerExchange version easier.

CAPT_XTRA

Specifies the path to a local directory on the UNIX system where you want to store CDC extraction maps.

```
CAPT_XTRA=path/directory
```

For the *path* variable, enter the path and directory name for storing extraction maps.

Default is the PowerExchange installation directory.

CMDNODE

The service name and connection information for a PowerExchange Listener or PowerExchange Logger for Linux, UNIX, and Windows process that is the target of pwxcmd commands.

Note: In this scenario, the PowerExchange Listener and PowerExchange Logger run in background mode. To issue commands to background processes, you must use the pwxcmd program. However, you can also use pwxcmd to issue commands in foreground mode or from a script file. For more information, see the *PowerExchange Command Reference*.

For this scenario, define CMDNODE statements in the DBMOVER configuration files on the UNIX and Windows machines so that you can issue pwxcmd commands from either machine.

Syntax:

```
CMDNODE=(service_name
        ,{CONDENSE|LISTENER}
        ,host_name
        ,connect_port
)
```

Parameters:

- *service_name*. A user-defined name for the command-handling service that is used for the PowerExchange process to which you want to send pwxcmd commands. For consistency, enter a name that matches the service name in the corresponding SVCNODE statement in the DBMOVER configuration file on the UNIX machine.
- {CONDENSE|LISTENER}. The service type. Enter CONDENSE for the PowerExchange Logger for Linux, UNIX, and Windows, or enter LISTENER for the PowerExchange Listener. No default is provided.
- *host_name*. The host name or IP address of the machine where the PowerExchange Listener or PowerExchange Logger runs. For this scenario, enter the host name or IP address of the UNIX machine.
- *port_number*. The port number on which the command-handling service for the PowerExchange Listener or Logger process listens for pwxcmd commands. Enter the port number that matches the port number in the corresponding SVCNODE statement in the DBMOVER configuration file on the UNIX machine.

If you specify a CMDNODE statement, also specify a corresponding SVCNODE statement.

No default value is provided.

LISTENER

Defines the TCP/IP port on which a named PowerExchange Listener process listens for work requests.

Syntax for this scenario:

```
LISTENER=({listener_node|node1}
          ,TCPIP
          ,{port|2480}
)
```

Parameters for this scenario:

- {*listener_node*|node1}. The node name of the TCP/IP port on which the PowerExchange Listener process listens. You specify this listener node name in the dtllst command when you start the PowerExchange Listener on the UNIX system.
- TCPIP. The communications protocol. TCPIP is the only valid option.
- {*port*|2480}. The TCP/IP port that the PowerExchange Listener uses to listen for work requests. Valid values are from 1 through 65535. Default is 2480.

LOGPATH

Specifies a unique path and directory name for PowerExchange message log files. Use this parameter to create message log files in a directory that is separate from your current working directory so that you can find PowerExchange messages more easily.

```
LOGPATH=path/directory
```

Default is the current working directory.

ORACLE_CAPTURE_TYPE

Specifies the type of PowerExchange Oracle CDC solution that is in use for the PowerExchange installation. In this scenario, enter D for PowerExchange Express CDC for Oracle.

Syntax:

```
ORACLE_CAPTURE_TYPE=D
```

Tip: To ensure consistent behavior, define the ORACLE_CAPTURE_TYPE statement on all systems that are involved in Oracle CDC processing, including the system where the CAPI_CONNECTION statements are defined.

ORACLEID

Specifies Oracle SID, database name, and connection information for the Oracle source database.

Syntax for this scenario:

```
ORACLEID=(collection_id
          ,oracle_db
         [,source_connect_string]
         [,capture_connect_string]
         )
```

Parameters for this scenario:

- *collection_id*. A user-defined name for this ORACLEID statement. This value must match the ORACOLL value in the ORAD CAPI_CONNECTION statement, the collection ID in the registration group that contains the source tables, and the DBID value in the PowerExchange Logger pwxccl configuration file. Maximum length is eight characters.
- *oracle_db*. The name of the Oracle database that contains the source tables registered for change data capture.
- *source_connect_string*. An Oracle connection string, defined in TNS, that the PowerExchange Navigator uses to connect to the Oracle source database. This connect string must be defined in the Oracle Client tnsnames.ora file on the UNIX source system.
- *capture_connect_string*. An Oracle connection string, defined in TNS, that the PowerExchange Logger uses to connect to the Oracle source database. This connect string must be defined in the Oracle Client tnsnames.ora file on the UNIX source system.

  Tip: Informatica recommends that you set the following environment variables, whenever possible, to enable connection to the Oracle database without using the *capture_connect_string* parameter and SQL*Net:

  - ORACLE_HOME
  - ORACLE_SID
  - PATH
  - LIBPATH or LD_LIBRARY_PATH

  Bypassing SQL*Net improves PowerExchange Logger performance, even when the PowerExchange Logger is running on the same machine as the Oracle source database.

Note: If you have multiple Oracle databases and capture changes from a database other than the default database, you must specify both the *source_connect_string* and *capture_connect_string* parameters.

You can specify up to 20 ORACLEID statements in the DBMOVER configuration file.

SVCNODE

The TCP/IP port on which a PowerExchange Listener or PowerExchange Logger for Linux, UNIX, and Windows process listens for commands. Define a SVCNODE statement for each PowerExchange Listener or PowerExchange Logger service to which you want to send pwxcmd commands.

Syntax:

```
SVCNODE=(service_name,port_number)
```

Parameters:

- *service_name* is one of the following node names:
  - For the PowerExchange Listener, the node name that is specified in the LISTENER statement.
  - For the PowerExchange Logger, the service name that is specified in the CONDENSENAME statement of the pwxccl.cfg file.
- *port_number*. An available port number on which the PowerExchange Listener or PowerExchange Logger listens for pwxcmd commands. This port number must be different from the one specified in the LISTENER statement.

If you specify a SVCNODE statement, also specify a corresponding CMDNODE statement.

TRACING

Enables PowerExchange to use alternative message log files for each PowerExchange process instead of the detail.log file for all PowerExchange processes. PowerExchange creates the alternative log files in a user-defined subdirectory of the LOGPATH directory. When an alternative log file becomes full, PowerExchange automatically switches to another log file. This statement can help you find messages for a PowerExchange process more easily and prevent errors related to the message logs becoming full.

Syntax for this scenario:

```
TRACING=(PFX=prefix
        [,APPEND={N|Y}]
        [,FILENUM={number_of_files|5}]
        [,FLUSH={flush_interval|99}]
        [,RECLEN={record_length|80}]
)
```

Parameters for this scenario:

- PFX. The name of an existing subdirectory of the LOGPATH directory in which to store alternative log files. In the subdirectory, the log file names vary based on the type of PowerExchange task that created the files.
- APPEND. Indicates how PowerExchange writes messages to existing log files after a restart. Use the default value of Y to have PowerExchange append new log records to the log file that was last used. Enter N to have PowerExchange overwrite the oldest log file.
- FILENUM. The number of dynamically allocated alternative log files. When a log file becomes full, PowerExchange switches to the oldest alternative log file and starts overwriting its contents. Valid values are 1 through 99. Default is 5.
- FLUSH. The number of records that PowerExchange collects before flushing them to the log file on disk. PowerExchange must periodically flush log records to recover from out-of-space conditions. Low flush values can increase I/O on the log file. Valid values are 1 through 99. Default is 99.
- RECLEN. The record length that PowerExchange uses to write log records to the log file. PowerExchange writes the log record on multiple lines if the message length exceeds the record length. Valid values are 80 through 255 characters. Default is 80.

For information about all DBMOVER statements, see the *PowerExchange Reference Manual*.

Example DBMOVER Configuration File on UNIX

The following example DBMOVER configuration file includes required and recommended statements for PowerExchange Express CDC for Oracle:

```
LISTENER=(pwxlst,TCPIP,2480)
NODE=(local,TCPIP,127.0.0.1,2480)
NODE=(pwxlst,TCPIP,127.0.0.1,2480)
NODE=(default,TCPIP,x,2480)
```

```
/*
CAPT_PATH=/Informatica/PowerExchange10.4.0/capture
CAPT_XTRA=/Informatica/PowerExchange10.4.0/capture/xtracmaps
LOGPATH=/Informatica/PowerExchange10.4.0/capture/logs
TRACING=/PFX=altlogs,APPEND=Y,FILENUM=5,FLUSH=99,RECLEN=80)
/*
/* The ORACLE_CAPTURE_TYPE statement specifies the Oracle CDC type.
/* Enter D for PowerExchange Express CDC for Oracle.
ORACLE_CAPTURE_TYPE=D
/*
/* Define an ORACLEID statement for each Oracle instance involved in CDC.
ORACLEID=(ORAD1DB,ORAD01,orad01.world)
/*
/* Use the ORAD CAPI_CONNECTION statement for PowerExchange Express CDC for Oracle.
/* Do not also specify ORCL CAPI_CONNECTION in this file.
CAPI_CONNECTION=(NAME=CAPORA3,TYPE=(ORAD,ORACOLL=ORAD1DB,
PARMFILE=/Informatica/PowerExchange10.4.0/capture/pwxorad.cfg))
/*
/* Define a CAPX CAPI_CONNECTION statement if you use the PowerExchange Logger
/* and continuous extraction mode.
CAPI_CONNECTION=(NAME=CAPX01,TYPE=(CAPX,DFLTINST=ORAD1DB))
/*
CAPI_CONN_NAME=CAPORA3
/*
/* To use pwxcmd to issue commands to PowerExchange Listener and Logger
/* processes, include the following statements:
SVCNODE=(pwxlst,8500)
SVCNODE=(pwxccl_svc,8501)
CMDNODE=(pwxlst,LISTENER,unix_host1,8500)
CMDNODE=(pwxccl_svc,CONDENSE,unix_host1,8501)
/*
```

Customizing the DBMOVER Configuration Files on the Windows and PowerCenter Integration Service Machines

To customize the DBMOVER configuration files on the Windows machine that hosts the PowerExchange Navigator and PowerCenter Client and on the PowerCenter Integration Service machine, add NODE statements that point to the UNIX machine that contains the Oracle source tables. PowerExchange uses these NODE statements to connect to the UNIX machine to read Oracle change data or metadata.

Note: You do not need to define a LISTENER statement in these DBMOVER files. The PowerExchange Listener runs only on the Oracle source machine to listen for SQL requests for data or metadata.

Example DBMOVER Configuration File Statements on the Windows and PowerCenter Integration Service Machines

In the DBMOVER configuration files on the Windows and PowerCenter Integration Service machines, add a NODE statement that points to the Oracle source system. Optionally, add CMDNODE statements to be able to issue pwxcmd commands to the PowerExchange Listener and PowerExchange Logger processes that run on the UNIX machine.

Example statements for the DBMOVER configuration file on each machine:

```
NODE=(ora_unix1,TCPIP,unix_host1,2480)
CMDNODE=(pwxlst,LISTENER,unix_host1,8500)
CMDNODE=(pwxccl_svc,CONDENSE,unix_host1,8501)
```

For more information about issuing pwxcmd commands and the command syntax, see the *PowerExchange Command Reference*.

Start the PowerExchange Listener

On the UNIX machine that contains the Oracle source tables, start the PowerExchange Listener.

To start the PowerExchange Listener in background mode and run it continuously, enter the following command from the directory where dtllist resides:

```
nohup dtllst node_name [config=PWX_installation/dbmover.cfg] [license=PWX_installation/
license.key] &
```

Parameters:

- **nohup**. Causes the PowerExchange Listener to run continuously, even if the session is disconnected or the user logs out.
- *node_name*. Specifies the value of the first positional parameter in the LISTENER statement in your dbmover.cfg file on the UNIX machine.
- **config**. Specifies the location of the DBMOVER configuration file. Required only if you placed the file in a directory other than the PowerExchange installation directory, as recommended.
- **license**. Specifies the location of the license.key file. Required only if you placed the file in a directory other than the PowerExchange installation directory, as recommended.
- **&**. Runs the PowerExchange Listener in background mode.

For this scenario, enter the following command:

```
nohup dtllst pwxlst config=Informatica/PowerExchange10.4.0/dbmover.cfg
license=Informatica/PowerExchange10.4.0/license.key &
```

Note: For more information, see the *PowerExchange Command Reference*.

Create PowerExchange Capture Registrations and Extraction Maps

For PowerExchange to capture changes, you must register the source database and each source table for change data capture (CDC).

First create the registration group to define the source location, source database type, a collection identifier, and, optionally, the user ID and password for accessing the source data. You assign a user-defined name to the group. PowerExchange adds an extraction group and application group of the same name.

To each registration group, you add one or more capture registrations. A capture registration defines a source table and the table columns for which to capture changes. You assign a user-defined name to the capture registration. Registration information is stored in the CCT file.

When you finish defining a capture registration, the PowerExchange Navigator creates a corresponding extraction map. You can use the extraction map to perform a database row test to verify access to source data. You can also import the extraction map into PowerCenter to create a source definition that can be added to a PowerCenter mapping.

PowerExchange assigns a registration tag name to both the capture registration and the generated extraction map for a source table. PowerExchange uses the registration tag name to identify the change records of interest for CDC processing. The PowerExchange Logger for Linux, UNIX, and Windows stores the registration tag name with each change record in its log files. For Oracle, the registration tag name has the following format:

```
ORAinstanceregistration_namen
```

Where:

- *instance* is the collection identifier that you specified in the registration group.
- *reg_name* is the registration name that you assigned to the capture registration.
- *n* is a number.

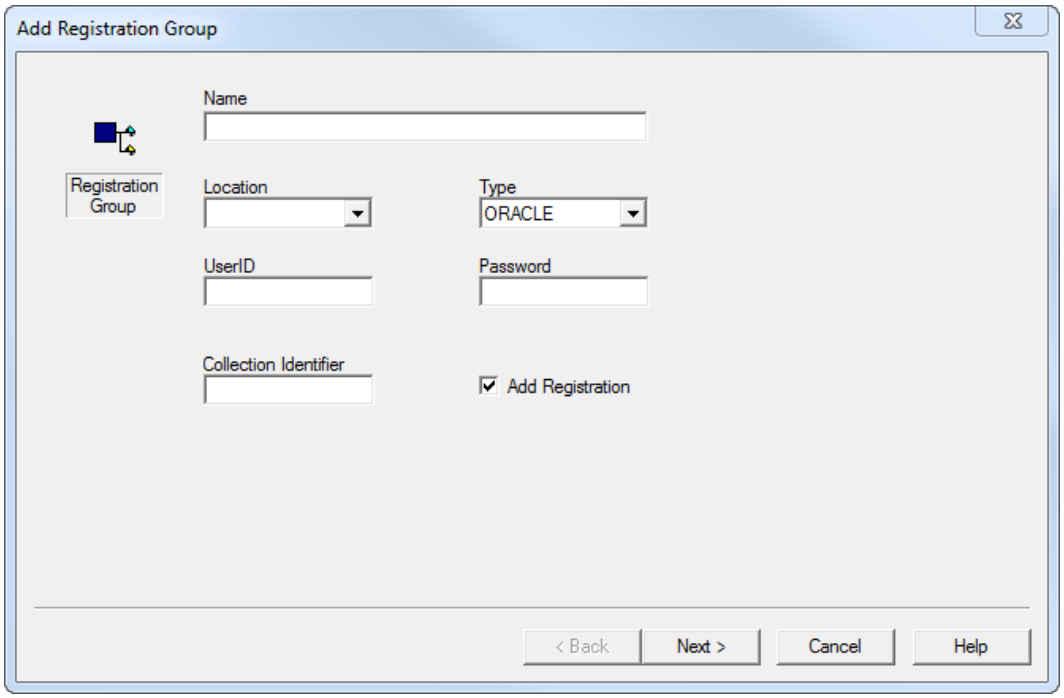
Creating Capture Registrations

In the PowerExchange Navigator, create a registration group and then add a capture registration for each Oracle source table to the group. When you finish a registration, PowerExchange automatically generates a corresponding extraction map.

A capture registration defines a source table from which to capture change data, the supplemental log group name for the table, and the registration status.

1. To start the PowerExchange Navigator on the Windows client, double-click **DTLUI.exe** in the root PowerExchange installation directory.
2. Click **Add** > **Registration Group**.

   The **Add Registration Group** page appears.



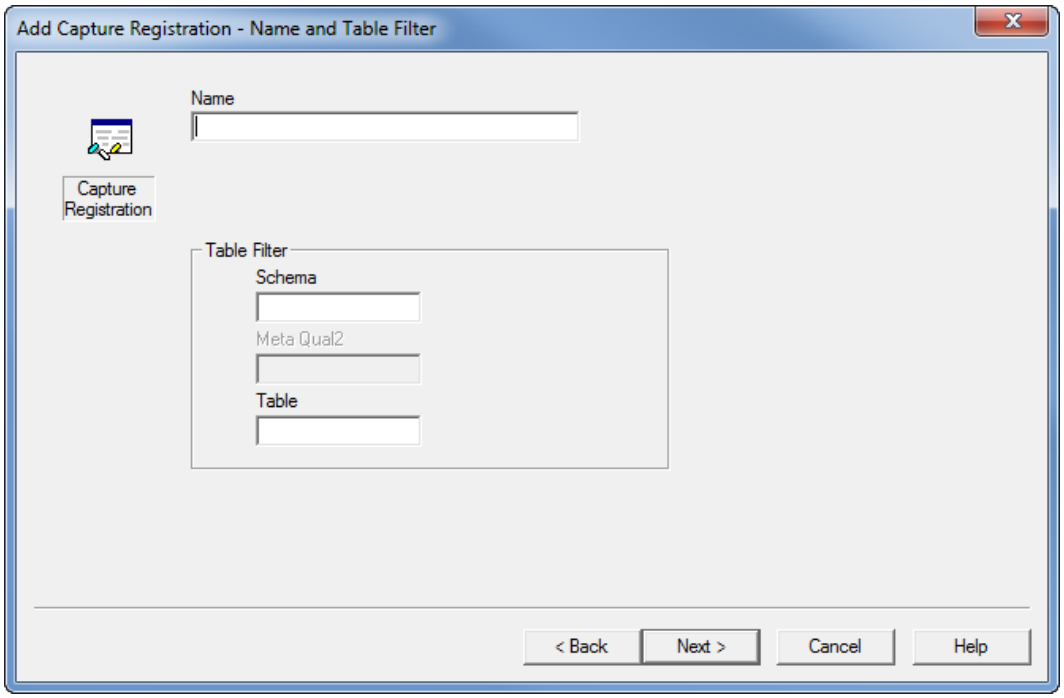
3. Enter information in the following fields:

Field	Description	Sample Value
Name	Enter a descriptive name for the registration group. Maximum length is 16 characters.	REG1OP
Location	Enter the user-defined node name for the UNIX machine that contains the Oracle source tables, as specified in a NODE statement in the local dbmover.cfg file on the Windows machine. The PowerExchange Navigator uses this information to connect to the UNIX machine.	ora_unix1
Type	Select ORACLE.	ORACLE
User ID	Enter a database user ID that has sufficient privileges to access the Oracle source tables for registration. This value should match the CAPTURE_NODE_UID value in the pwxccl.cfg file.	ORACAPTL1

Field	Description	Sample Value
Password	Enter the database password that is associated with the specified user ID. This value should match the CAPTURE_NODE_PWD or CAPTURE_NODE_EPWD value in the pwxccl.cfg file.	Oracaptl1
Collection identifier	Enter the name that you specified for the first parameter in the ORACLEID statement of the dbmover.cfg file. This value should also match the ORACOLL value in the ORCL CAPI_CONNECTION statement and the DBID value in the pwxccl.cfg file.	ORAD1DB
Add Registration	Select this option to add capture registrations for the Oracle source tables to the registration group.	-

Then click **Next**.

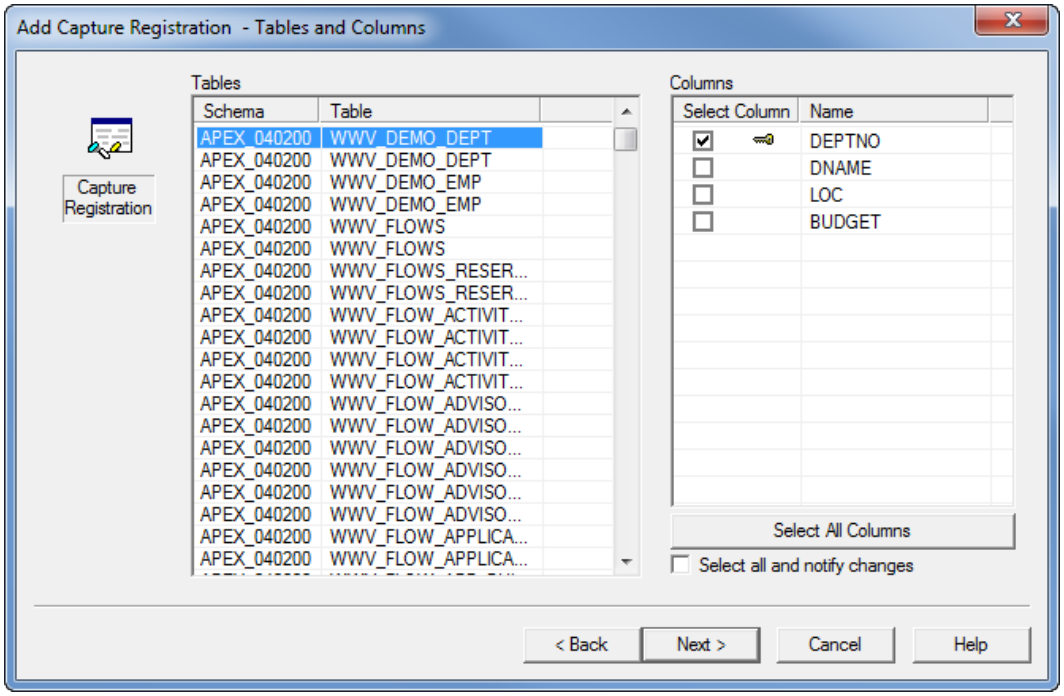
The **Add Capture Registration – Name and Table Filter** dialog box appears.



4. Enter information in the following fields:
   - In the **Name** field, enter a descriptive name for the registration in lowercase. This name can be up to eight alphanumeric characters in length and must begin with a letter.
   - Optionally, In the **Schema Name** and **Table Name** fields, enter a schema name mask or a table name mask or both to filter the list of tables to register. Use the asterisk (*) wildcard anywhere in one of these masks to represent one or more characters.

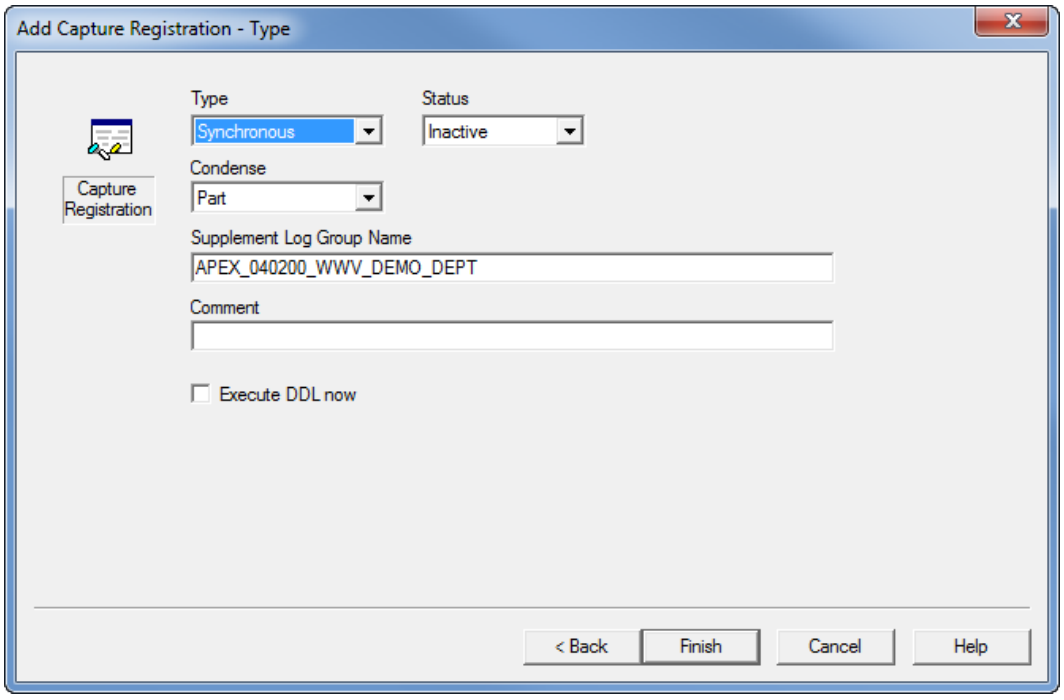
   Then click **Next**.

The **Add Capture Registration – Tables and Columns** page appears and lists all tables that match any filter criteria you entered.



5. In the **Tables** list, double-click a table that you want to register. Then in the **Columns** list, select the table columns for which to capture changes. Click **Next** to continue.

   The **Add Capture Registration – Type** page appears.



6. Enter information in the following fields:

Field	Description
Type	Select **Synchronous**.
Status	Select **Active** to set the registration status to active. If you do not want to activate the registration for capture processing now, you can set the status later.
Condense	Select **Part** to use the PowerExchange Logger.
Supplemental Log Group Name	Enter a supplemental log group name, or accept the default name, which has the format of *schema_tablename*. A supplemental log group must be defined for each table from which you want to capture Oracle change data. When you click **Finish**, the wizard will prompt you to save the DDL for creating the supplement log group to a file.
Execute DDL now	Select this option to have PowerExchange run the DDL for creating the supplemental log group when you click **Finish**. Do not select this option if you do not have the privileges that are required to create supplement log groups on the Oracle table. Instead, give the file with generated DDL to your Oracle DBA.

   Then click **Finish**.

   The **Save Supplement Log Group SQL to File** dialog box appears.

7. Browse to the directory to which you want to save the SQL file that contains the DDL for creating the supplemental log group and click **Save**.

   If you selected the **Execute DDL now** option, the DDL runs.

The PowerExchange Navigator displays the capture registration.

For more information, see the *PowerExchange Navigator User Guide*.

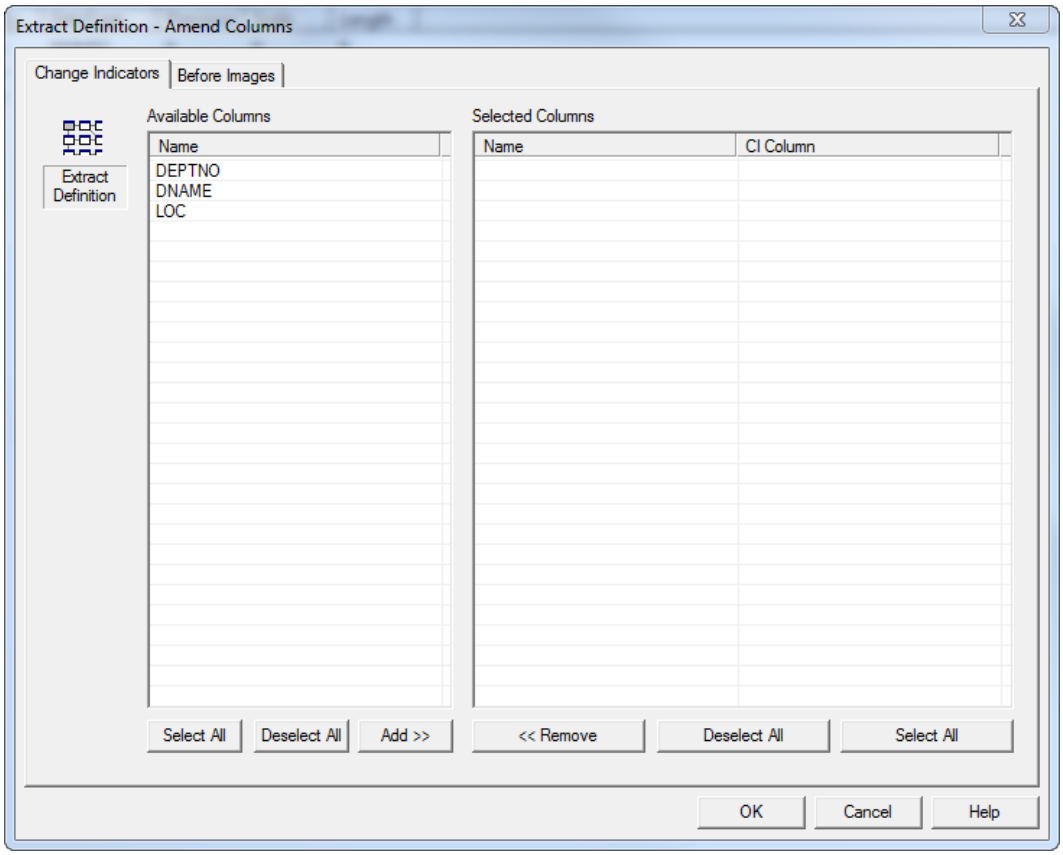
Adding Change Indicator and Before Image Columns to the Extraction Maps (Optional)

In the PowerExchange Navigator, you can optionally edit the extraction maps for registered source tables to add PowerExchange-generated change indicator (DTL_CI) and before image (DTL_BI) columns.

A DTL_CI column indicates if UPDATEs caused data in the column to change. Based on this information, you can perform some processing, such as capture data only for the columns for which the indicator is Y. A DTL_BI column stores the before image of data in a column that was updated. If you do not add DTL_BI columns, only the after image is captured.

1. In the PowerExchange Navigator, open an extraction map for a registered Oracle source table.
2. Right-click in the **Extraction Definition** window and select **Amend Change Indicator/Before Image Extensions**.

The **Extract Definition - Amend Columns** dialog box appears.



3. To add DTL_CI and DTL_BI columns, perform the following actions:
   a. On the **Change Indicators** tab, select the columns in the **Available Columns** list to which you want to add a DTL_CI column. Then click **Add** to add the column to the **Selected Columns** list.
   b. On the **Before Images** tab, select the columns In the **Available Columns** list to which you want to add a DTL_BI column. Then click **Add**.
   c. Click **OK**.

   The DTL__CI_column_name and DTL_BI_column_name columns appear in the **Extraction Definition** window.

   Note: You can remove columns from an extraction map by clearing clear the column check boxes. Changes to the column data are still captured but not extracted.
4. Save the extraction map.

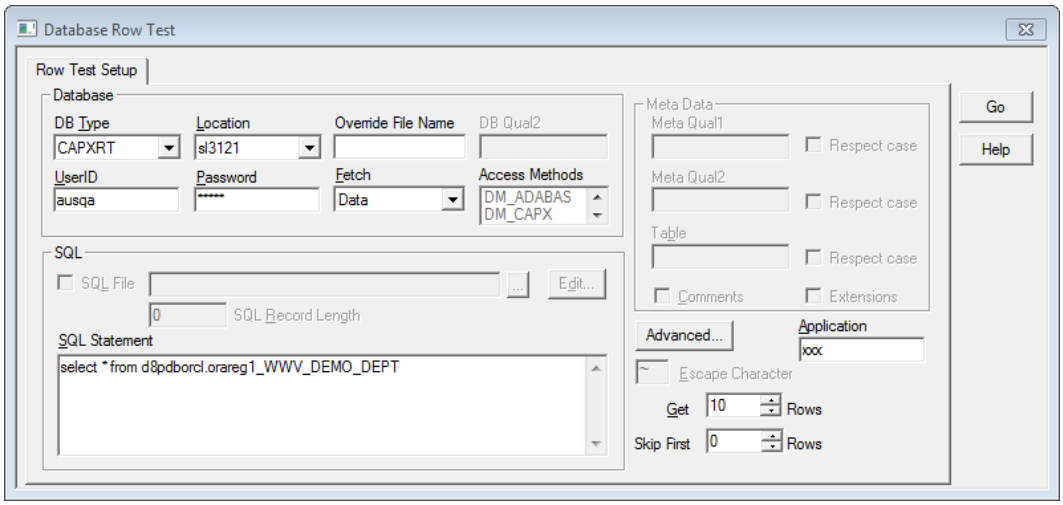
For more information, see the *PowerExchange Navigator User Guide*.

Performing a Database Row Test on an Extraction Map

In the PowerExchange Navigator, you can perform a database row test on an extraction map to verify that PowerExchange can use it to access PowerExchange Logger log files and capture change data in near real time.

1. In the PowerExchange Navigator, select an extraction map for a registered Oracle source table.
2. Click **File** > **Database Row Test**.

The **Database Row Test** dialog box appears.



3. In the **DB_Type** list, select **CAPXRT**.
4. In the **Application** field, enter any value. This value is required but serves no function for a row test.
5. In the **Location**, **User ID**, and **Password** fields, verify that the values match the values that you specified for the related registration group.
6. In the **Fetch** list, verify that **Data** is selected.
7. Verify that the **SQL Statement** box displays a SELECT statement for the correct extraction map.
8. Click **Go**.

   When the row test completes, the **Database Row Test Output** area at the bottom displays the columns values in the change records retrieved from the PowerExchange Logger log files. If the window is blank, change data might not yet have been written to the log files. A blank window does not necessarily indicate an error.

Tip: After the PowerExchange Logger is running and has logged changes, you can perform a database row test to verify that change data is being logged to the PowerExchange Logger log files. Before running the row test, you must click **Advanced**. Then in the **CAPI Connection Name** field in the **CAPXRT Advanced Parameters** dialog box, enter the name of the CAPX CAPI_CONNECTION statement, CAPX01, to use the continuous extraction connection.

For more information, see the *PowerExchange Navigator User Guide*.

Materialize the Target Tables

You must create the target tables and initially populate them with data from the Oracle source tables before starting the CDC session.

First stop updates to the source tables. Then use PowerCenter, PowerExchange bulk data movement, or another tool to populate the target tables with data.

If you use PowerCenter, you can create a mapplet and re-use it in two workflows: one workflow for the initial target materialization, and another workflow that uses a PWX Oracle CDC connection for CDC. The workflows will have different source connection types and session properties. If you use PowerExchange bulk data movement, you need to create separate PowerCenter mappings for bulk data movement and CDC. For more information, see the *PowerExchange Interfaces for PowerCenter* publication.

Cold Start the PowerExchange Logger

On the UNIX machine with the Oracle source tables, cold start the PowerExchange Logger to begin capturing changes to the Logger log files. A cold start is required when you start the PowerExchange Logger for the first time.

If the RESTART_TOKEN and SEQUENCE_TOKEN parameters are not defined in the pwxccl.cfg file, the PowerExchange Logger starts from the current end-of-log (EOL).

To cold start the PowerExchange Logger, enter the following command from the directory where pwxccl program resides:

```
pwxccl coldstart=y
```

If you store the pwxccl.cfg file in a directory other than the PowerExchange installation directory, you must also include the cs option to point to the file:

```
pwxccl coldstart=y cs=directory\pwxccl.cfg
```

After the PowerExchange Logger has been cold started once, you can run it continuously in the background by entering the following command:

```
nohup pwxccl cs=directory\pwxccl.cfg 2>&1 < /dev/null &
```

Because the coldstart=y parameter is omitted, the PowerExchange Logger warm starts. A warm start uses the restart and sequence tokens in the CDCT file to resume CDC processing. You can perform a warm start only if you have run the PowerExchange Logger previously and have a recent CDCT file for the database instance.

For more information, see the *PowerExchange Command Reference*.

After you cold start the PowerExchange Logger, allow users to resume writing changes to the Oracle source tables.

Configure PowerCenter to Process PowerExchange Change Data

PowerExchange works with PowerCenter to transmit change data to targets.

To configure PowerCenter to retrieve change data from PowerExchange Logger log files and transmit the data to targets, you must perform the following tasks in PowerCenter:

1. Import the extraction maps for the source tables to create source definitions that can be included in the mapping.
2. Create the target table definitions.
3. Create the mapping.
4. Configure a PWX Oracle CDC Real Time application connection.
5. Create a workflow and session.

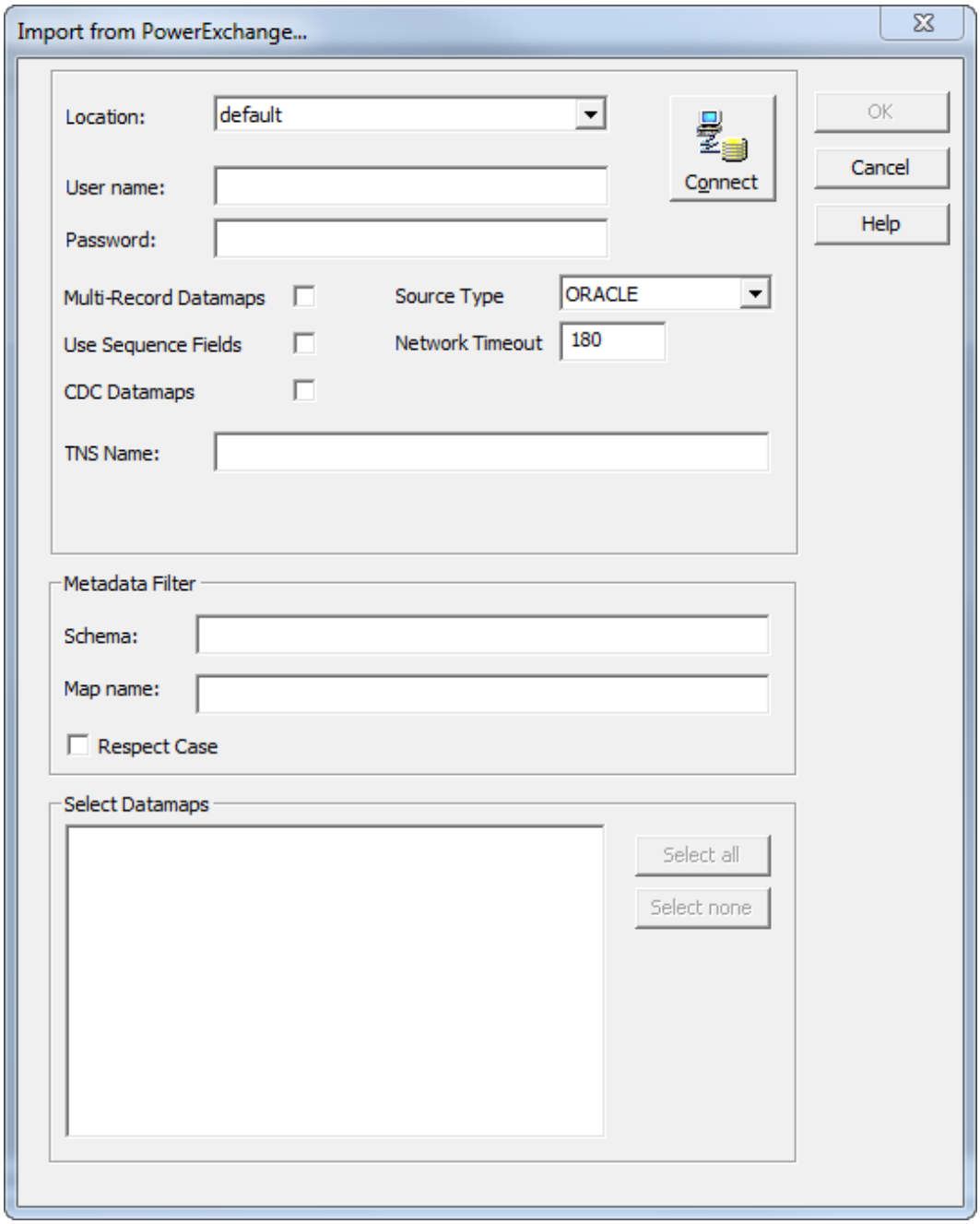
Importing Extraction Maps for Source Tables

To create an Oracle source definition in PowerCenter, import the PowerExchange extraction map for each source table.

Although PowerExchange can use the PowerExchange Listener to read table metadata directly from the Oracle database, Informatica recommends that you import the extraction map instead. The extraction map provides the additional DTL__ columns that PowerExchange generates, including the DTL__CAPXACTION column with the change type (INSERT, UPDATE, or DELETE). Also, the extraction map includes any DTL__CI and DTL__BI columns that you added for change indicators and before images.

1. Start PowerCenter Designer.
2. In the Source Analyzer, click **Sources** > **Import from PowerExchange**.

The **Import from PowerExchange** dialog box appears.



3. Enter the following information:

Field	Description
Location	Enter the name that you specified as the first parameter in the LISTENER statement of the dbmover.cfg file on the UNIX machine where the Oracle source tables reside. In this scenario, enter pwxlst.
User Name	Enter a database user name. For this scenario, enter ORACAPTL1.
Password	Enter the password associated with the user name. For this scenario, enter Oracaptl1.
Source Type	Select **ORACLE**.
CDC Datamaps	Select this check box to list extraction maps in the **Select Datamaps** box.
Schema	Optionally, enter a schema name mask to filter the extraction maps that will be returned. Permitted wildcards: Use an asterisk (*) to represent one or more characters anywhere in the name. Use a question mark (?) for a single character.
Map name	Optionally, enter a map name mask to filter the extraction maps.

4. Click **Connect**.

   The **Select Datamaps** box lists the extraction maps that match your criteria.

5. Select one or more extraction maps to import.
6. Click **OK**.

   In the Repository Navigator, the Oracle source table definitions appear under ORACLE_*location*.

Importing Target Table Definitions

In PowerCenter, import metadata for your target tables.

The method that you use depends on your target type.

- If you have a target type that PowerCenter supports, in the PoweCenter Target Designer, use the **Import from Database** option to import metadata for the target. Then, define a PowerCenter connection object for the target.
- If you have a target type that PowerExchange supports but PowerCenter does not support, such as a Db2 for z/OS target, use the **Import from PowerExchange** option to import metadata for the target from PowerExchange. Then, define a PowerExchange Client for PowerCenter (PWXPC) connection or a PowerExchange ODBC connection.

For more information, see *PowerExchange Interfaces for PowerCenter* and *PowerCenter Designer Guide*.

Creating a Mapping

Create a mapping that includes all of the Oracle source definitions and the target definitions.

In this scenario, you create a single mapping. Group source processing is used for CDC automatically to enhance throughput and reduce resource consumption. With group source processing, PowerExchange, in conjunction with PWXPC, reads change data for all of the Oracle source tables from the PowerExchange Logger log files in a single pass. The source tables must be on the same Oracle instance.

Note: If you have a large number of source tables in a mapping, for example more than 25, memory might be insufficient to edit the mapping in Designer. In this case, consider creating multiple mappings.

To create a mapping, use Mapping Designer. Add your source and target definitions and link them. For more information, see the *PowerCenter Designer Guide*.

Tip: Informatica recommends that you use the prefix "m_" for mapping names, for example, m_ora01_to_db2_cdc.

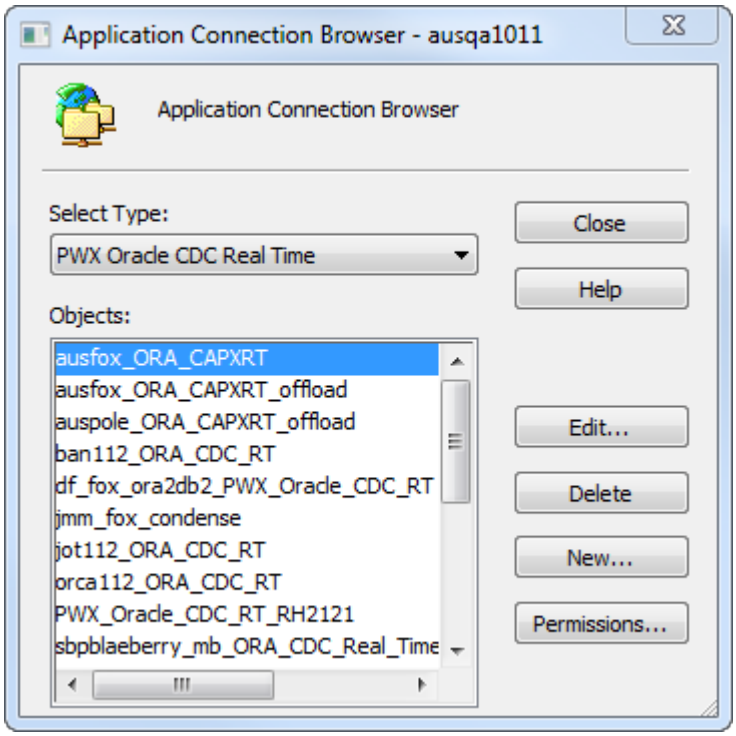
Configuring an Application Connection for the Oracle Sources

Configure a generic application connection for Oracle sources. For continuous extraction mode, you must use a connection type of PWX Oracle CDC Real Time.

Note: You can re-use the generic connection object by customizing connection attributes when you create the CDC session in PowerCenter.

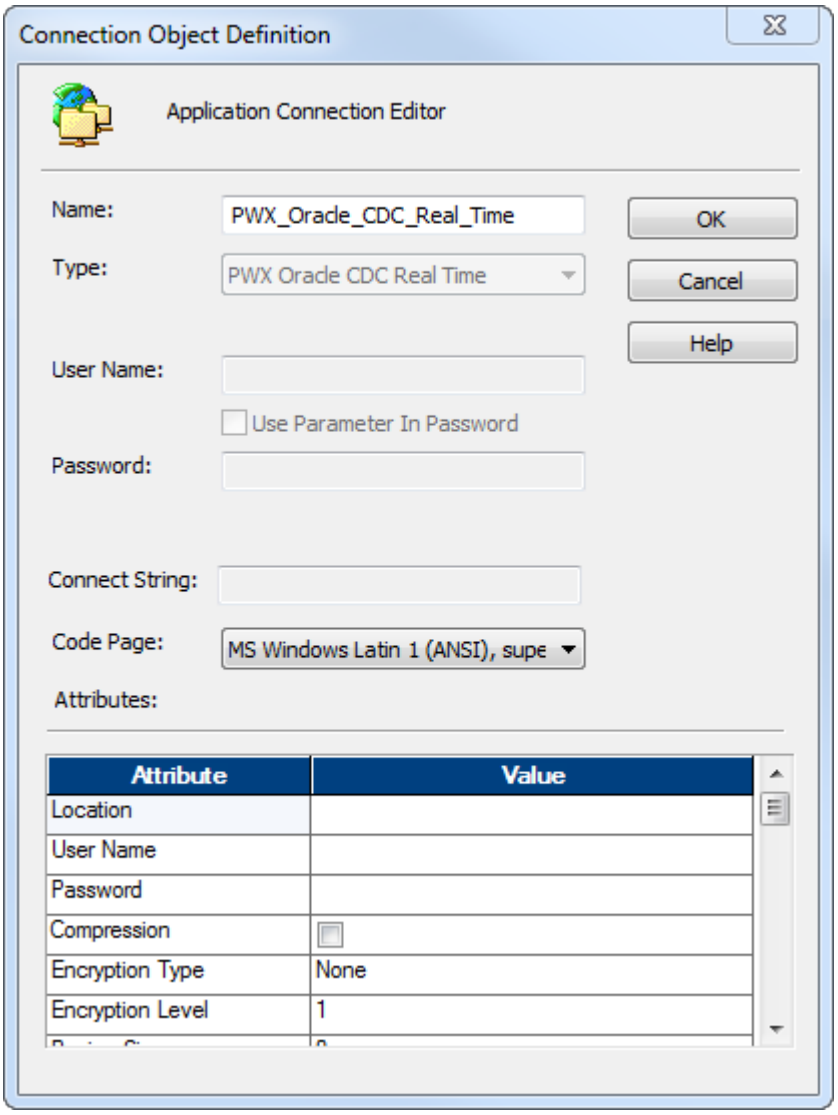
1. In the Workflow Manager, click **Connections** > **Application**.

   The **Application Connection Browser** dialog box appears.



2. In the **Select Type** box, select **PWX Oracle CDC Real Time**.
3. Click **New**.

The **Connection Object Definition** dialog box appears.



4. Enter information for the following attributes:

Attribute	Value
Name	Enter a name for this connection object.
CAPI Connection Name Override	To configure continuous extraction, you must provide the name of the CAPX CAPI_CONNECTION statement in the CAPI Connection name override attribute for the PWX Oracle CDC Real Time connection. In continuous extraction mode, PowerExchange reads change data continuously from open and closed PowerExchange Logger log files in near real time. In real-time extraction mode, PowerExchange reads change data directly from the database log files in near real time, on an ongoing basis. When the PowerExchange Listener receives an extraction request, it pulls the change data from the log files and transmits the data to PowerCenter for extraction.

Attribute	Value
Code Page	Select the code page for the PowerCenter Integration Service to use to extract source data. In Unicode mode, you must select UTF-8 for Oracle CDC connections.
Location	Enter the node name for the UNIX machine where the PowerExchange Listener that stored the extraction maps for the Oracle source tables resides. This node name must be specified in a NODE statement in the dbmover.cfg file on the Integration Service machine. For this scenario, use ora_unix1.
User Name	Enter the Oracle user name. For this scenario, use ORACAPTL1.
Password	Enter the password for the user name. For this scenario, use Oracaptl1.
Image Type	Select **AI**, even though the CAPT_IMAGE parameter in the PowerExchange Logger pwxccl.cfg file specifies BA. In this scenario, both before images and after images are captured but only the after images are extracted and written to targets.
Restart Token File Folder	Accept the default value of $PMRootDir/Restart as the restart token file directory. Only with this default can PWXPC automatically create the directory.
Number of Runs to Keep Restart Token File	Enter the maximum number of backup copies of the restart token file to keep. Default is 0. Specify a value other than 0 or 1 to retain more than a single pair of initialization and termination restart token files. For this scenario, enter 5.

5. Click **OK**.
6. Click **Close**.

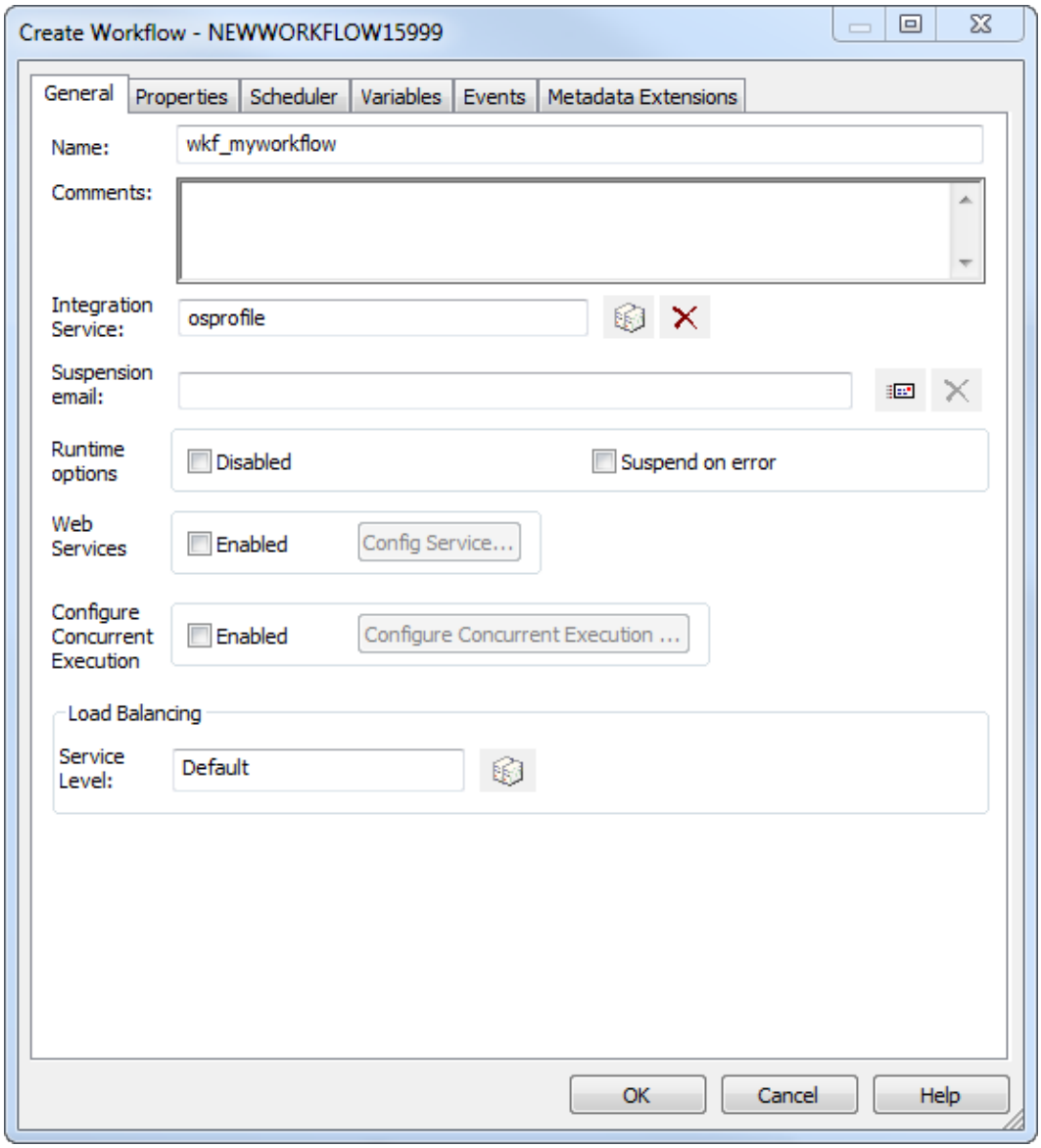
For more information about defining PWX CDC application connections, see the *PowerExchange Interfaces for PowerCenter* publication.

Creating a Workflow and Session

Create a workflow and session task to provide instructions to the PowerCenter Integration Service for extracting and applying change data. As part of defining the session, customize connection object attributes for the session. When you start the workflow, change data that has been captured to the PowerExchange Logger log files begins flowing to the targets.

1. Start the Workflow Manager.
2. Open Workflow Designer.
3. Click **Workflows** > **Create**.

The **Create Workflow** dialog box appears.

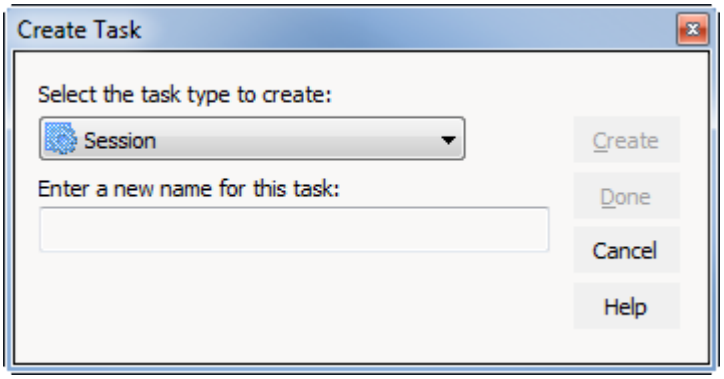


4. Enter a workflow name and click **OK**.

   Tip: Informatica recommends that you use the prefix “wkf_” for workflow names, for example, wkf_ora01_to_db2_cdc.

5. Open Task Developer.
6. Click **Tasks** > **Create**.

The **Create Task** dialog box appears.

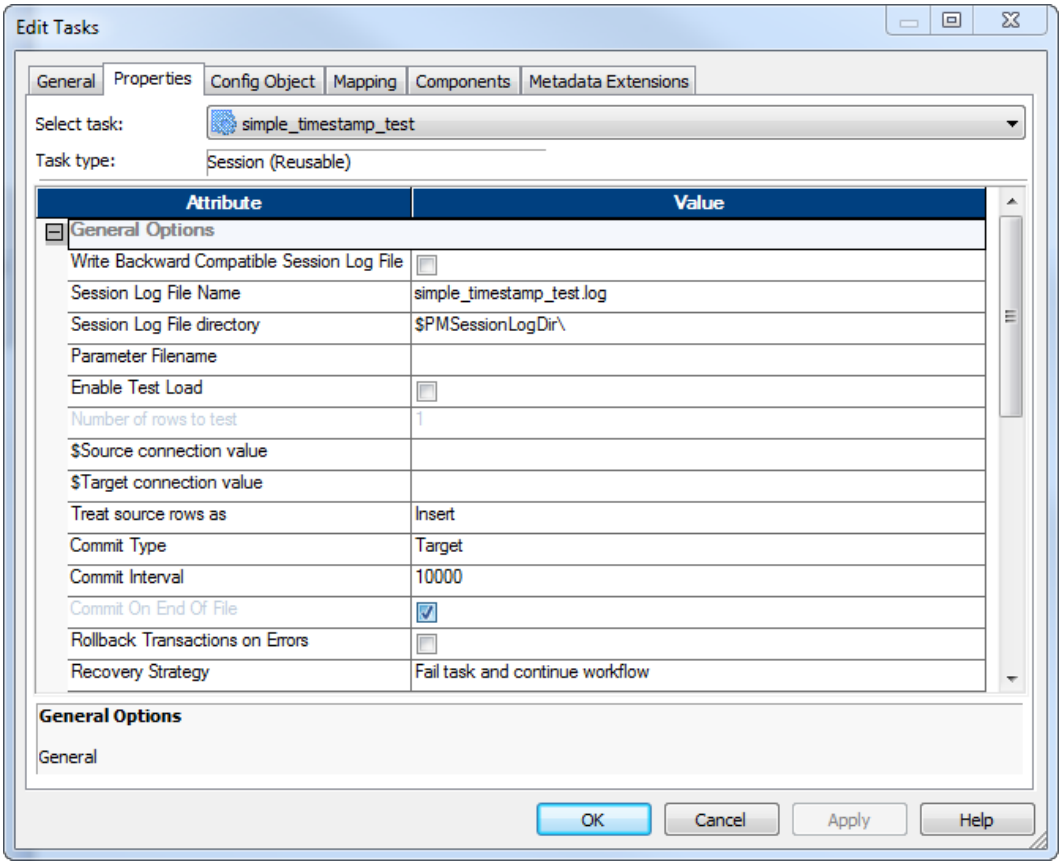


7. Select **Session** as the task type and enter a name for the task. Then select the mapping that you want to associate with the session.

   Tip: Informatica recommends that you use the prefix "s_" for session names, for example, s_ora01_to_db2_cdc.

8. Click **Create** and then click **Done**.
9. To customize session attributes and options, double-click the new session task.

   The **Edit Tasks** dialog box appears.



10. Set the attributes and options that are required or recommended for CDC, and then click **OK**.

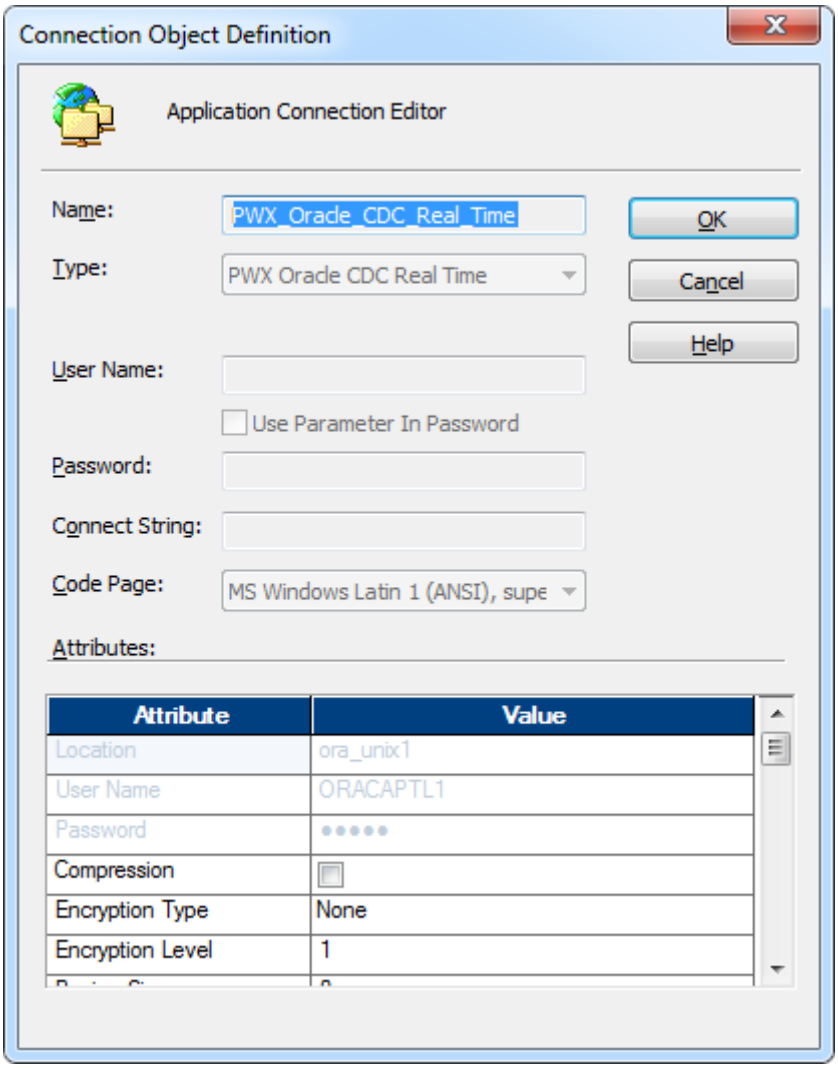
The following table describes these attributes and options by tab:

Attribute or Option	Tab	Value
Commit Type	Properties	Select **Source**. This setting is required to reset **Commit On End Of File**.
Commit On End of File	Properties	Clear this check box. Otherwise, the Integration Service performs a commit when the session ends, after PWXPC has already committed restart tokens. This situation can cause duplicate data to occur after restart.
Recovery Strategy	Properties	Select **Resume from last checkpoint**. This setting is required for PWXPC and the Integration Service to properly resume extraction processing and target loads after a session stops.
DTM buffer size	Properties	Enter the amount of buffer memory to allocate to the session from the Data Transformation Manager (DTM) process for reading, writing, and transforming data. By default, the Integration Service determines the DTM buffer size at run time. Often, the result of the automatic calculation is not appropriate for a CDC session, and you need to specify a buffer size. The DTM buffer size is related to the **Default buffer block size** property on the **Config Object** tab. The number of buffer blocks in a session is equal to the `DTM buffer size / buffer block size`. The minimum number of buffer blocks must be greater than the total number of sources and targets. For optimal CDC performance, set the **DTM buffer size** and **Default buffer block size** properties to values that result in a large number of small blocks. Informatica recommends the following settings: - For **DTM buffer size**, specify 512 MB. - For **Default buffer block size**, specify 32 KB. **Note:** For both of these session properties, the default unit type of bytes is used unless you append KB, MB, or GB. Do not set these session properties to **Auto**. For more information, see the *PowerCenter Advanced Workflow Guide*.

Attribute or Option	Tab	Value
Stop on errors	Config Object	Enter the number of non-fatal errors that the Integration Service can encounter before it stops the session. If the session writes data to relational targets on Linux, UNIX, or Windows or to PowerExchange nonrelational targets on z/OS, enter 0. With this setting, non-fatal errors do not cause the session to stop. The session continues processing data from all sources, even if one has an error. To determine if non-fatal errors occurred, periodically review the contents of the target reject file. If the session writes data to PowerExchange Db2 for i5/OS or Db2 for z/OS targets, enter a non-zero value to cause the workflow to abort when the specified number of errors is met. Otherwise, the workflow continues running and might log an excessive number of error messages. Use the following criteria to set this value: - Enter 1 if you expect the data to be clean and the target database *not* to get duplicate-key or referential-integrity errors. - Otherwise, enter a large value based on the data volume that the workflow processes and the number rows that can be reprocessed, for example, 10000. The default value is set in the configuration that you used to create the session. The default_session_config sets **Stop on Errors** to 0.
Connections: Type and Value	Mapping	Select the first Oracle source table in the left pane. Then, under **Connections**, select **Application** in the **Type** field. In the corresponding **Value** field, select the generic CDC Real Time connection object that you created. For Oracle source tables other than the first table, select **None** in the **Type** field. **Note:** In the next step, you customize connection attributes for the first source table and then apply the connection type and attributes to all of the other tables that have a connection type of **None**.

11. On the **Mapping** tab, the first Oracle source table in the left pane. Then under **Connections** in the right pane, click the pencil icon next to the **Value** field.

The **Connection Object Definition** dialog box appears.



12. Customize the following Oracle source connection attributes for the session:

Attribute	Value
Application Name	Enter a unique application name. Maximum length is 20 characters. **Warning:** If you use the same application name for different CDC sessions, data loss or damage to data integrity might occur.
Restart Token File Name	Enter a unique file name for the restart token file.

Attribute	Value
UOW Count	Enter the number of units of work (UOWs) that PWXPC processes before it sends a commit request to PowerCenter to commit data to all targets in the CDC session. Valid values: **-1 or 0**. Disables this attribute. ***n***. Specifies the number of UOWs that PWXPC processes before sending a commit request. Default is 1. If you have many small UOWs, you can use the **UOW Count** or **Minimum Rows Per commit** attribute or both to create a consistent flow of UOWs that can be committed to the target in batches of approximately the same size. You can also control commits that occur at UOW boundaries based on time by specifying the **Real-time Flush Latency** attribute. Specify the **UOW Count** or **Real-time Flush Latency** attribute or both. If you specify both, a commit occurs when either limit is reached.
Idle Time	Enter the number of seconds the PowerExchange Listener remains idle after reaching the end of the change log, as indicated by the message PWX-09967, before returning an end-of-file (EOF). Valid values: - **-1**. An EOF is not returned. The session runs continuously. - **0**. An EOF is returned at the end of log. The session ends successfully. - ***n***. An EOF is returned after no data has been processed for *n* seconds. The session ends. Default is -1. Typically, the default value is used for real-time sessions.
Real-time Flush Latency in milliseconds	Enter the number of milliseconds that must elapse before PWXPC sends a commit request to PowerCenter. When this period elapses, PWXPC continues to read the changes in the current UOW until it reaches the end of the UOW. Then PWXPC sends a commit request to PowerCenter so that the data can be committed to the targets. Valid values: - **-1**. Disables this attribute. - **0 to 86400**. The number of milliseconds that must elapse before PWXPC sends a commit request to PowerCenter. If you specify a value from 0 through 2000, PWXPC uses 2000 milliseconds (2 seconds). You can use this attribute to ensure that data is committed to the target periodically when transaction rates are low and the **UOW Count** or **Maximum Rows Per commit** limit is not being met within the desired maximum latency period. You can also control commits that occur at UOW boundaries based on number of UOWs by specifying the **UOW Count** attribute. You can specify the **UOW Count** or **Real-time Flush Latency** attribute or both. If you specify both, a commit occurs when either limit is reached.
Maximum Rows Per commit	Enter the maximum number of change records that PWXPC processes before it sends a commit request to PowerCenter to commit data to all targets in the CDC session. PWXPC does not wait for a UOW boundary to commit the change data. Default is 0, which disables this attribute. Use this attribute to perform subpacket commits for very large UOWs.

Attribute	Value
Minimum Rows Per commit	Enter the minimum number of change records that PowerExchange Listener must read from the change stream before it passes a commit record to PWXPC. Before reaching this minimum, the PowerExchange Listener passes only change records to PWXPC and discards any commit records. Default is 0, which disables this attribute. If you have many small UOWs, you can use the **Minimum Rows Per commit** or **UOW Count** attribute or both to create a consistent flow of UOWs that can be committed to the target in batches of approximately the same size.
CAPI Connection Name Override	Enter the name of the CAPX CAPI_CONNECTION statement that you defined in the dbmover.cfg on the UNIX machine with the Oracle source. For this scenario, enter CAPX01.

When you are done, click **OK** to return to the **Edit Tasks** dialog box. Then, right-click the **Application** field and select **Apply Connection Attributes** > **To all Instances** to apply the connection object attributes to the other Oracle source tables.

13. On the **Mapping** tab, select the first target table in the left pane and customize attributes.

    The attributes that you set depend on your target type. You should at least set the following attributes:

    - For the **Target load type** attribute, select **Normal**.
    - For the **Table Name Prefix** attribute, enter the schema or owner name of the target table. Apply this attribute to all instances.

14. Link the Start task and session task.
15. Save the workflow.

For more information about creating workflows and sessions, see the *PowerCenter Workflow Basics Guide* and *PowerExchange Interfaces for PowerCenter*.

Configuring the Restart Token File

The restart token file provides restart tokens for specifying the point in the change stream from which to start new CDC sessions. You can create a restart token file that sets specific restart tokens for each Oracle source table or that generates restart tokens that represent the current end of the change stream. Alternatively, you can let PWXPC create a restart token file that contains null restart tokens.

In this scenario, let PWXPC create a restart token file that contains null restart tokens. With null restart tokens, change extraction processing begins with the first PowerExchange Logger log file.

During session initialization, PWXPC uses the name specified in the **Restart Token File Name** attribute to create an empty restart token file, if one does not exist. The restart token file name is one of the following values, in order of precedence: the **Application Name** value in the connection attributes, the workflow name, or the **Restart Token File Name** value in the connection attributes. By default, the workflow name is used. However, the workflow name might not result in a unique restart token file name. The restart token file name must be unique for each CDC session. Otherwise, session failures and potential data loss might occur. To ensure a unique name, verify that you entered an application name value in the source connection attributes.

PWXPC creates the restart token file in the directory that is specified in the **Restart Token File Folder** attribute of the source CDC connection attributes. If this attribute contains the default value of $PMRootDir/Restart, PWXPC also automatically creates this directory, if it does not exist.

During change data extraction processing, PWXPC updates restart tokens and stores the updated token values in the PM_REC_STATE table in the target database. You can use the restart token file to override the restart tokens in the PM_REC_STATE table, if necessary, for a warm start.

Important: Review the restart and recovery information in *PowerExchange Interfaces for PowerCenter*. You must understand basic concepts about PowerExchange CDC restart and recovery to prevent change data loss and to preserve target data integrity.

Performing a Cold Start of the CDC Workflow

Cold start the workflow to begin moving captured change data from the PowerExchange Logger log files to the target tables. When you perform a cold start, PWXPC reads the restart tokens from the restart token file only.

In Workflow Manager, right-click the workflow in the **Respository Navigator** and select **Cold Start Workflow**.

The CDC session starts. How long it runs depends on the **Idle Time** setting in the connection attributes. If you accepted the default value of -1, the session runs continuously until it is stopped or interrupted.

For more information about cold starts, warm starts, and recovery processing, see *PowerExchange Interfaces for PowerCenter*.

Additional Information

For more information about PowerExchange CDC for Oracle, see the following resources:

- PowerExchange documentation:
  - *PowerExchange CDC Guide for Linux, UNIX, and Windows*: Describes configuration of the PowerExchange Logger and PowerExchange Express CDC for Oracle on Linux, UNIX, or Windows.
  - *PowerExchange Interfaces for PowerCenter*. Covers installing PWXPC and configuring mappings, PWX Oracle CDC Real Time connections, and CDC session attributes. Also describes PWXPC restart and recovery processing.
  - *PowerExchange Reference Manual*. Describes all dbmover.cfg configuration file parameters and PowerExchange security configuration.
  - *PowerExchange Command Reference*: Describes PowerExchange Listener and PowerExchange Logger commands, including the pwxcmd commands.

  To access these PowerExchange publications, browse the Informatica Knowledge Base (KB) at https://kb.informatica.com/_layouts/ProductDocumentation/Page/ProductDocumentSearch.aspx.
- How-To articles:
  - "Using Heartbeat Tables to Determine the Latency of Change Data Flow"
  - "Creating a PowerExchange Capture Registration and Extraction Map for an Oracle Source"
  - "Migrating to PowerExchange Express CDC for Oracle"

  To access these how-to articles, search for "how-to library" from https://kb.informatica.com and then select the **PowerExchange** product filter.
- Knowledge Base (KB) articles: Numerous KB articles related to PowerExchange Express CDC for Oracle are available. To access these KB articles, search for PowerExchange Express CDC from https://kb.informatica.com and select **Knowledge Base** as the source.

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